Hamiltonian Neural Networks for Solar System

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Abstract

This paper investigates a structurally constrained, physics-informed geometric deep learning framework designed to model a 10-body configuration of the Solar System over astronomical timescales up to 400 years. Conventional Hamiltonian Neural Networks (HNNs) frequently exhibit optimization stagnation and localized numerical instabilities—typically leading to unphysical inner planet ejections within 10 to 50 years—due to the high dynamic range of physical magnitudes across celestial bodies. To address these baseline characteristics, we employ a Vector-wise Multiscale V2 normalization scheme alongside an explicit Kinetic-Potential (T-V) energy separation via the Plausible HNN (P-HNN). While dense P-HNN architectures extend the sustainable orbital lifespan up to 200 years, scaling the framework over a 4,000-year dataset to encompass Pluto's full orbital threshold (~ 248 years) introduces parameter inflation and computational bottlenecks during training. To manage this overhead, we develop the Block (Sparse) Plausible HNN (BLK-P-HNN), which incorporates a Topology Adaptor to isolate position inputs into 45 independent sub-manifolds via batch matrix multiplications. This structural sparsity compresses the model's total weight footprint 400-fold, from 88.6 MB down to 200 kB. Empirical evaluation across the 400-year integration window demonstrates a distinct cost-performance trade-off: while the dense P-HNN maintains a lower average tracking error and isotropic dispersion, the compressed BLK-P-HNN successfully maintains a gravitationally bound state for the entire 400-year duration, albeit at the expense of introducing subtle localized orbital variations and increased dispersion along the Z-axis. As a parallel approach, we introduce the Separable Latent Linear Hamiltonian Neural Network (SLL-HNN). This model uses latent linear mappings to separate kinetic and potential energies. The implementation of this explicit energy separation within a two-layer dense network was evaluated over a 400-year integration window, maintaining dynamically stable planetary orbits up to a strict 200-year physical limit. Performance-wise, the SLL-HNN demonstrates comparable initial stability to the baseline P-HNN and BLK-P-HNN architectures; however, it accumulates higher numerical drift and lacks their parameter efficiency. Consequently, this variant establishes its role strictly as an architectural proof-of-concept. Together, our findings show that enforcing physical constraints

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either through structural sparsity or explicit energy separation effectively preserves long-term stability in multi-body simulations.

1. Introduction

The long-term orbital prediction of multi-body gravitational configurations remains a foundational yet computationally formidable challenge in celestial mechanics [1, 2]. Historically, deterministic trajectories of complex stellar and planetary systems have been mapped utilizing high-order symplectic integrators, such as the Wisdom-Holman or Yoshida schemes, to preserve the structural geometric invariants of Hamiltonian systems [3]. While these classical numerical algorithms deliver exceptional fidelity, integrating chaotic or high-dimensional systems over astronomical timescales demands an intractable computational burden, rendering rapid, iterative long-term simulations prohibitively expensive [4]. To alleviate this operational overhead, the recent paradigm of Physics-Informed Neural Networks (PINNs) has emerged as a promising alternative, seeking to leverage the universal function approximation capabilities of deep learning to bypass expensive step-by-step numerical integration [5].

A pivotal milestone in this domain was the inception of Hamiltonian Neural Networks (HNNs) by Greydanus et al. (2019), which mathematically enforce the conservation laws of symplectic geometry by embedding Hamilton's equations directly into the network's optimization objective [6]. To further ensure structure preservation, advanced variants such as symplectic networks [7] and controlled Hamiltonian ODE networks [8] have been proposed. However, the evaluation of these architectures has largely been restricted to low-dimensional toy problems (e.g., simple pendulums or non-chaotic few-body systems). Their capability remains unproven in realistic, high-dimensional multi-body environments, where complex orbital resonances and long-term chaotic behaviors—such as those found in actual solar system dynamics—frequently lead to significant numerical drift over extended horizons. When attempting to model a highly coupled 10-body configuration of the Solar System, standard dense HNN architectures encounter a severe High Dynamic Range (HDR) problem, characterized by extreme disparities in physical magnitudes—such as mass, distance, and orbital velocity—across inner and outer celestial bodies. Under these unmitigated HDR conditions, standard models suffer from severe gradient imbalances and optimization pathologies, as characterized in physics-informed frameworks [9]. Consequently, our preliminary investigations reveal that instead of preserving the long-term orbital stability intrinsic to the Solar System [10], these standard architectures produce critical numerical artifacts. Specifically, we observe a compounding geometric instability that culminates in a domino effect of unphysical inner planetary ejections within short integration windows of merely 10 to 50 years.

To systematically break this geometric barrier and mitigate multi-body trajectory drifts, this paper introduces the Plausible HNN (P-HNN) architecture alongside its structurally optimized version: the Block (Sparse) Plausible Hamiltonian Neural Network (BLK-P-HNN). Shifting entirely away from the generic black-box topologies used in standard HNNs, our proposed P-HNN architecture enforces strict structural inductive biases by executing an explicit Kinetic-Potential (T-V) energy separation and exploiting functional spatial symmetries via even activation functions. Combined with the Vector-wise Multiscale V2 normalization scheme designed to balance heterogeneous physical scales across celestial bodies, this baseline dense P-HNN successfully extends the system-level lifespan up to 200 years within unseen test domains, representing an immediate advancement over conventional HNN boundaries by preventing catastrophic orbital decay.

To manage these architectural constraints and investigate the balance between model efficiency and long-term macro-scale stability, we develop the BLK-P-HNN architecture. The defining structural element introduced in this framework is a Topology Adaptor that isolates position inputs into 45 independent planet-to-planet sub-manifolds via batch matrix multiplications. By systematically eliminating redundant cross-planet weight connections that do not correspond to physical pairwise interactions, the BLK-P-HNN implements a structured sparsity that reduces the model's total weight footprint 400-fold —from 88.6 MB down to 200 kB. Empirical tracking over a 400-year integration window reveals an informative performance trade-off: while the extreme parameter compression introduces subtle localized orbital variations and an increased error dispersion concentrated along the vertical Z-axis compared to the dense baseline, the sparse network successfully maintains a gravitationally bound state and prevents catastrophic multi-body planetary ejections or decays throughout the entire astronomical horizon.

As a parallel variant stemming from the baseline Plausible HNN (P-HNN), the Separable Latent Linear Hamiltonian Neural Network (SLL-HNN) architecture is introduced. This model explicitly bifurcates the computation of kinetic and potential energies through component-wise linear projections into a latent space. Its implementation directly integrates the Vector-wise Multiscale V2 normalization scheme to mitigate the high dynamic range (HDR) problem. This formulation operates without injecting artificial numerical stabilizers that alter the underlying physics of the system. At the optimization level, the network analysis corroborates that traditional odd activation functions induce severe numerical drifts and premature orbital ejections, reinforcing the imperative use of even functions seen in P-HNN and BLK-P-HNN architectures to preserve spatial symmetry. Contrary to the expectation that deeper networks are required, empirical evaluations demonstrate that a compact two-layer configuration optimally mitigates overfitting. This specific setup confines planetary trajectories within a bounded region. While this architecture trades the extreme parameter efficiency of the BLK-P-HNN for a denser 600 kB formulation, it serves as an independent proof-of-concept demonstrating that enforcing physical invariants via even activations maintains macro-scale stability, even within the constraints of strictly linear embeddings.

In summary, the primary contributions of this work are threefold:

1. We analyze the scalability boundaries of standard HNNs under high-dynamic-range constraints, and demonstrate through the proposed P-HNN framework how explicit Kinetic-Potential (T-V) separation serves to suppress catastrophic multi-body orbital decay.

2. We evaluate the BLK-P-HNN architecture governed by a batch-matrix-multiplication Topology Adaptor, providing a quantitative mapping of the trade-off between extreme parameter compression (400-fold reduction) and long-term macro-scale gravitational bound-state stability over a 400-year horizon.

3. We also introduce the Separable Latent Linear Hamiltonian Neural Network (SLL-HNN), which explicitly decouples kinetic and potential energies via latent linear mappings.

2. Data Preparation

The data set was synthetically generated using a 12th-order Yoshida symplectic integrator. This numerical scheme is essential to guarantee the long-term conservation of the geometric properties of the Hamiltonian phase space, avoiding artificial energy dissipation in the N-body problem simulation [18].

Section 2.1: Simulation Protocol and Initial Data

The initial conditions for the 10-body system (Sun, eight planets, and Pluto) were obtained from the NASA HORIZONS system. To ensure physical consistency, the phase space was transformed to the

Center of Mass (CM) reference frame, eliminating artificial translational drifts through the following corrections:

$$q'_{i,t0} = q_{i,t0} - \frac{\sum m_j q_{j,t0}}{\sum m_j}, \qquad p'_{i,t0} = p_{i,t0} - \frac{m_i}{\sum m_j} \sum p_{j,t0}$$

The integration used a gravitational constant $G = 4\pi^2 \mathrm{AU}^3 / (M_{\odot} \cdot \mathrm{year}^2)$ and a time step $\Delta t = 0.0005$ years, reaching a 10,000-year horizon $(20 \times 10^6$ states$)$. The computational details of this process, as well as the numerical coefficients of the integrator's decomposition, are documented in the project's public repository [16], serving as the high-precision database for the subsequent training of the model.

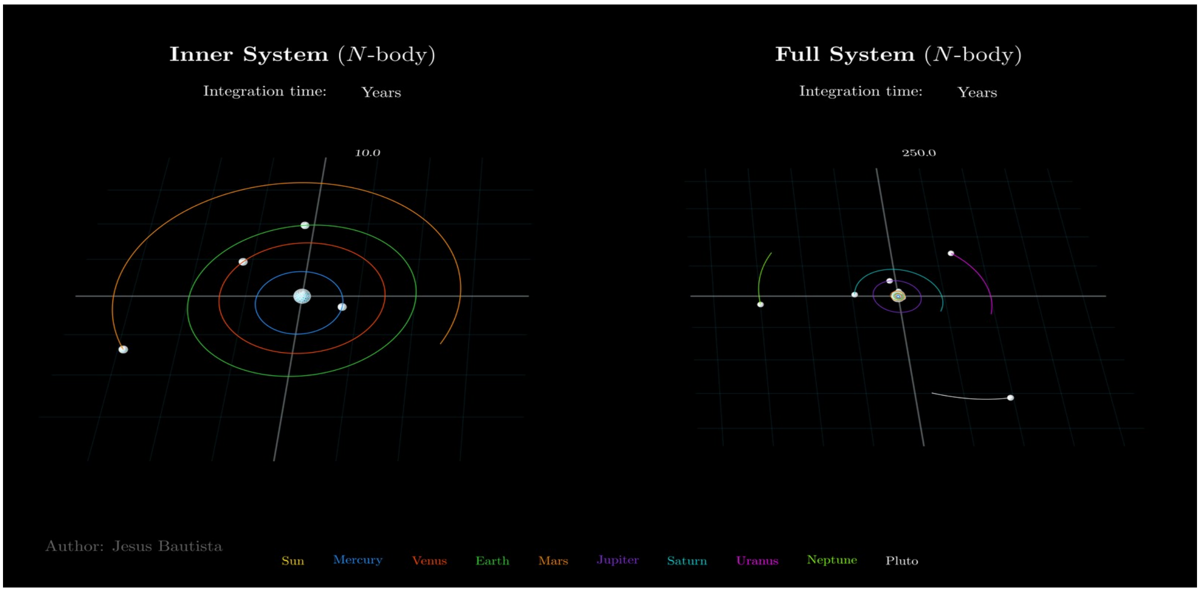


Figure 1: Orbital evolution of the system. *The left panel illustrates the inner planetary system over a 10-year horizon, while the right panel shows the full system configuration (including outer bodies) at 250 years, validating the dynamical consistency of the integration.*

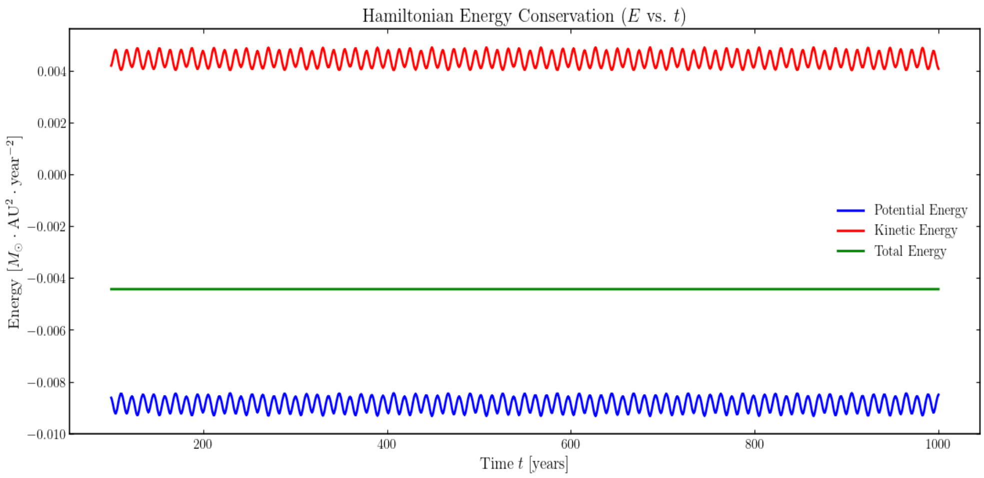


Figure 2: Hamiltonian stability. *The relative error of the system's energy is presented, monitored between year 100 and year 1,000. The bounded oscillation of the Hamiltonian demonstrates the preservation of the symplectic structure and the high numerical precision required for neural network training.*

At the dynamical level, the robustness of the higher-order integrator guarantees the long-term stability of the textured orbits in the phase space. As illustrated in **Figure 1**, even in the inner orbits of the planetary system (where relative velocities and orbital frequencies are significantly higher and prone to numerical degradation), the trajectories remain stable with virtually imperceptible perturbations or numerical noise.

This consistency is based on the energetic behavior of the system. **Figure 2** shows how the kinetic energy (represented in red) and potential energy (represented in blue) oscillate in a coupled and compensatory manner through a quasi-sinusoidal behavior. This harmonic transfer of energy translates into a practically horizontal total Hamiltonian (black line), visually demonstrating the time invariance of the system's energy. To quantify the precision of this conservation, **Figure 3** exhibits the residual error of the total energy, which remains bounded in infinitesimal magnitudes with peak values fluctuating on the order of $\left(1\times10^{-10}\%\right)$. This asymptotic behavior confirms that the data set constitutes a state space free of artificial dissipation, which is optimal for the geometric learning of the neural network.

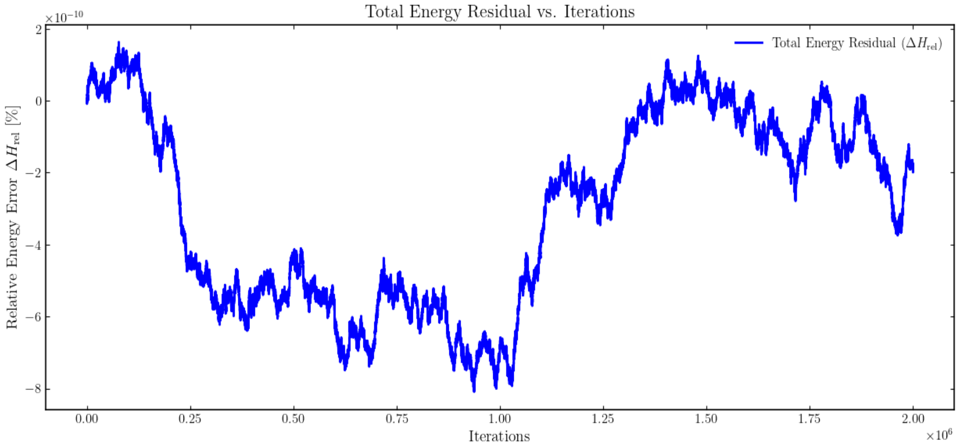


Figure 3: Residual error profile of the Hamiltonian. *The Percentage fluctuation of the total energy of the integrated symmetric system. The non-cumulative variations remain within an order of magnitude of* $(\approx 1\times 10^{-6}\%)$, *validating the symplectic conservation.*

3. Methodology

Section 3.1: Conventional HNN Architecture (Baseline)

The conventional HNN, denoted as the baseline model, follows the canonical design proposed by Greydanus et al. (2019). The architecture is designed to map a 60-dimensional state vector comprised of scaled positions $\hat{q}$ and momenta $\hat{p}$ for the 10 celestial bodies to a single scalar value representing the Hamiltonian in the scaled state space, $H(\hat{q},\hat{p})$.

Structural Specification:

- **Input Layer:** A 60-element tensor $(\hat{q}_0,\dots,\hat{q}_{29};\hat{p}_0,\dots,\hat{p}_{29})$.
- **Hidden Layers:** Two consecutive blocks of fully connected layers (nn.Linear) followed by the hyperbolic tangent (nn.Tanh) activation function. All linear layers include learnable bias terms by default.
- **Output Layer:** A final linear layer reducing the hidden features to a 1-dimensional scalar H.

Learning Mechanism:

The model is trained by embedding the symplectic structure directly into the $(\hat{q},\hat{p})$ state space. The network does not learn the coordinates directly; instead, it optimizes the Hamiltonian surface such that its partial derivatives satisfy the Hamilton's equations:

$$\frac{d\hat{p}}{dt}=-\frac{\partial H}{\partial \hat{q}}, \qquad \frac{d\hat{q}}{dt}=\frac{\partial H}{\partial \hat{p}} \tag{1}$$

By backpropagating through these gradients, the system ensures that the phase space flow remains on the manifold of constant energy, at least locally within the scaled domain.

Section 3.2: Plausible HNN Architecture (Bridge Model)

To address the limitations of the black-box approach, we developed the Plausible HNN, an intermediate architecture designed to validate the impact of explicit energy separation and functional symmetry as illustrated from **Figure 5** to **Figure 7**. This model serves as a crucial baseline for demonstrating the effectiveness of the Even Activation function before transitioning to the sparse block-based design.

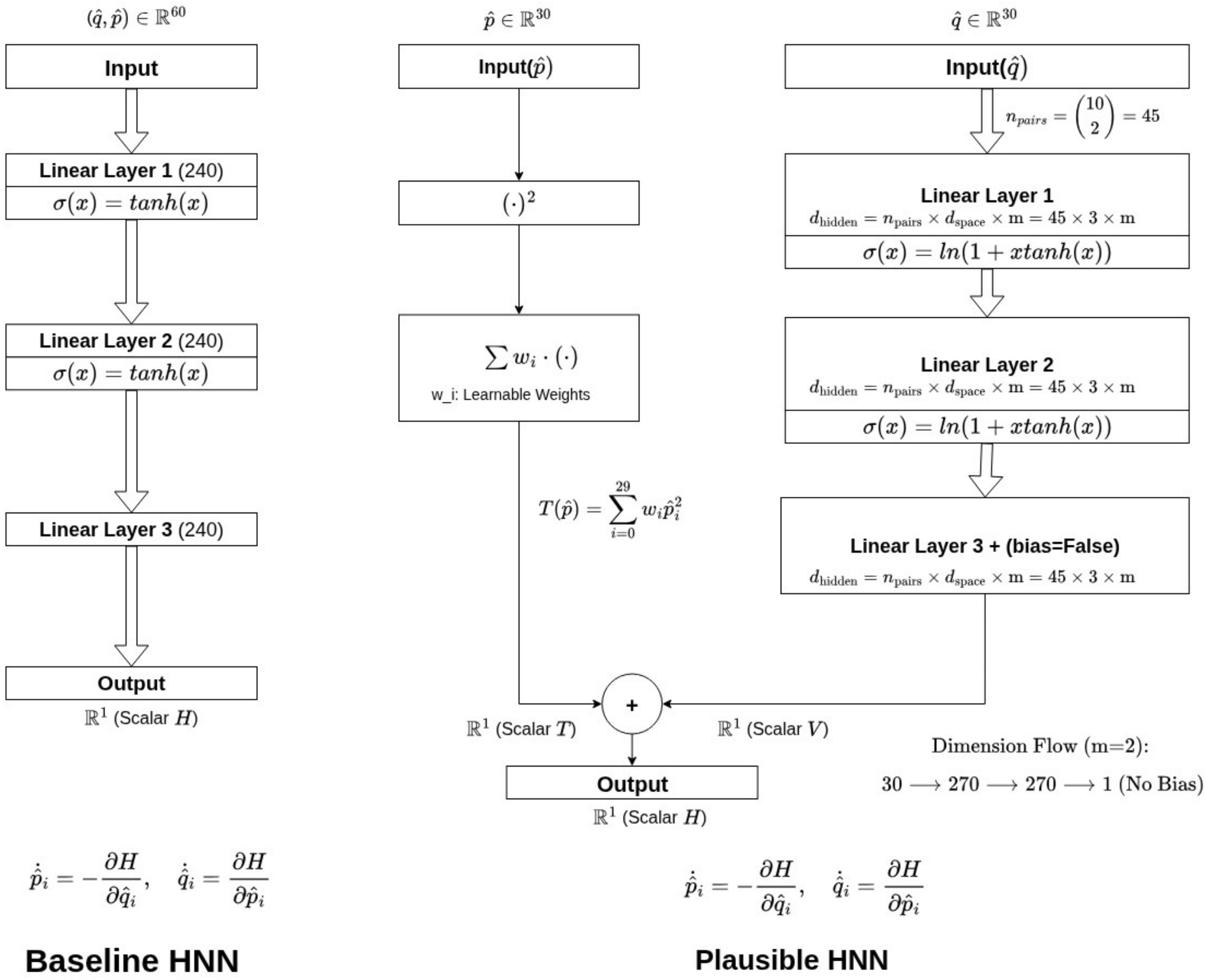


Figure 4: Architectural evolution of Hamiltonian Neural Networks (HNNs) for celestial dynamics. *(a) Conventional HNN mapping the global 60-dimensional scaled state vector via a standard dense black-box MLP; (b) Plausible HNN enforcing analytical prior via explicit Kinetic-Potential (T-V) energy separation and parity symmetry through custom Even Activation*

1. Kinetic-Potential Energy Separation (T-V Split)

Unlike the conventional HNN, which uses a single MLP to map the entire state vector to a scalar, the Plausible HNN strictly enforces the additive nature of the Hamiltonian: $H = T(\hat{p}) + V(\hat{q})$.

- **T-net (Analytical Prior):** The kinetic energy is modeled as a weighted quadratic sum of momenta, $T = \sum w \cdot \hat{p}^2$. By using a learnable parameter w for each dimension, the network captures the diagonal components of the kinetic energy matrix directly.

- **V-net (Dense MLP):** The potential energy is modeled using a Dense MLP that receives only the position vector $\hat{q}$.

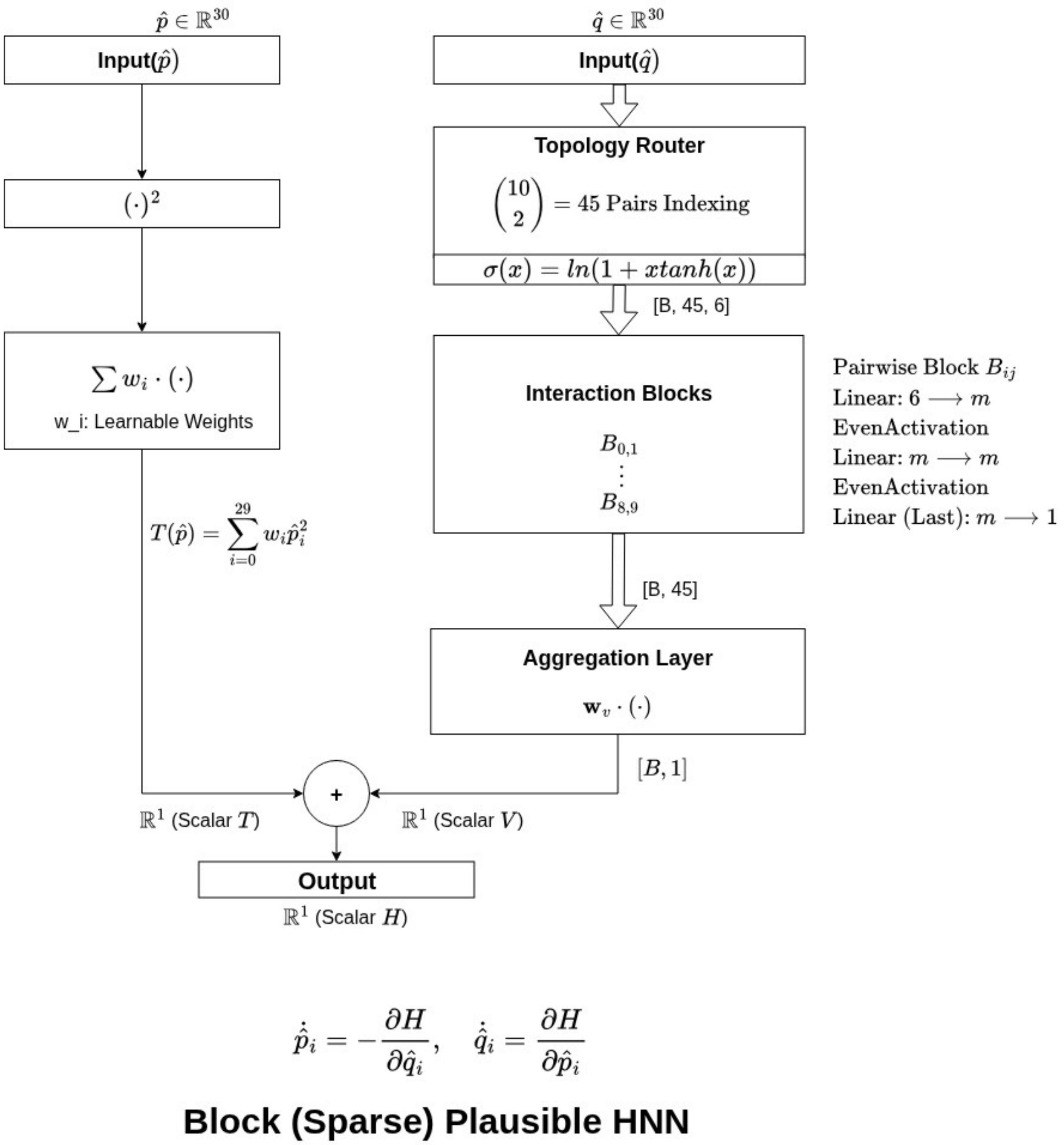


Figure 5: Block Plausible HNN hardcoding the pairwise topology to maximize parametric efficiency.

2. Symmetry Bias via Even Activation

The defining feature of this model is the introduction of a custom Even Activation function

$$f(x) = \ln(1 + x \cdot \tanh(x)) \tag{2}$$

This function is strictly symmetric ($f(x) = f(-x)$), ensuring that the learned potential field reflects the even parity often required for stable energy surfaces. This architectural bias prevents the model from learning unphysical odd-order gradients that often cause the rapid orbital collapse (divergence) observed in standard Tanh-based HNNs.

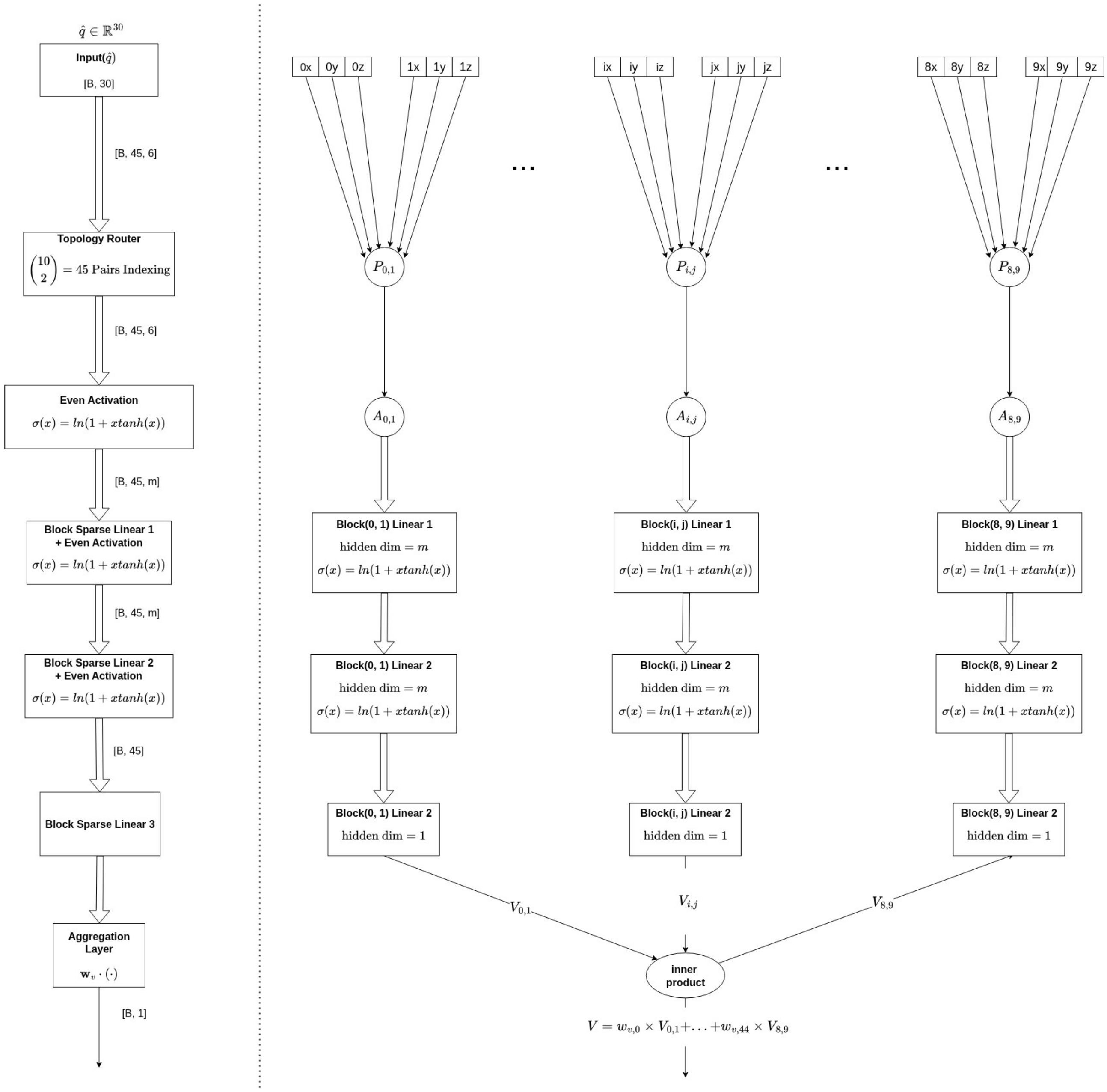


Figure 6: Detailed schematic of the Pairwise Topology Router and data slicing mechanism within the Block Plausible HNN. *The global position tensor* $\boldsymbol{X} \in \boldsymbol{R}^{\boldsymbol{B\times 30}}$ *is explicitly mapped into* $n_{\text{pairs}} = {}_{10}C_2 = 45$ *isolated interaction sub-manifolds (*$\boldsymbol{R}^{\boldsymbol{B\times 1\times 6}}$¿*) using a static indexing buffer. Each block extracts unique coordinates (e.g.,* $(0_x, 0_y, 0_z)$ *and* $(1_x, 1_y, 1_z)$ *for* $\text{Pair}_{0,1}$ *to enforce independent, parallel feature extraction via Batch Matrix Multiplication (BMM).*

Section 3.3: Block (Sparse) Plausible HNN Architecture

To achieve extreme parametric efficiency, we implemented a Block Sparse connectivity pattern that decomposes the global potential energy into independent pairwise contributions. The architecture is realized through the following specialized modules:

1. Pairwise Input Linear Adaptor (Data Routing)
The initial layer, Topology Adaptor, acts as a physical router. Instead of processing the entire 30-dimensional position tensor $\hat{q}$ globally, it utilizes a pre-computed indexing buffer to extract 45 unique celestial pairs ($_{10}C_2$).

- **Mechanism:** For each pair (i, j), it slices a 6-dimensional sub-tensor representing the 3D coordinates of both bodies.
- **Parallel Processing:** These pairs are processed in parallel using Batch Matrix Multiplication (BMM), ensuring that the initial linear transformation is applied independently to each interaction block.

2. Pairwise Hidden Block Layers (Modular Evolution)
The intermediate layers consist of `PairWiseHiddenBlockLinear` modules. Unlike conventional dense layers where every neuron connects to every neuron in the next layer, this module maintains 45 isolated weight manifolds.

- **Sparsity:** Information regarding the interaction of Earth-Mars, for instance, never mixes with Sun-Jupiter data within the hidden layers.
- **Efficiency:** This structural isolation reduces the parameter count from a $O(N^2)$ dense matrix to a block-diagonal-like structure, directly contributing to the reduction of the model size to ~ hundreds of kilo-Bytes.

3. Attention-based Potential Aggregation

The final scalar potential V is not a simple sum but a weighted aggregation.

- **Weighted Summation:** The network produces 45 individual scalar potentials V_{ij}. These are then combined via a learnable weight vector $\boldsymbol{w}_v$ (Attention-like mechanism): $V = \boldsymbol{w}_v^T \boldsymbol{V}_{\mathbf{pair}}$.
- **Physical Significance:** This allows the model to learn the relative significance (e.g., gravitational coupling strength) of each pair.

The implementation of these novel HNN architectures, Plausible HNN (P-HNN) and Block Sparse Plausible HNN (BLK-P-HNN) are available on the github [17].

Section 3.4: Separable Latent Linear Hamiltonian Neural Network

The proposed architecture, Separable Latent Hamiltonian Neural Network (SLL-HNN), is based on neural operator methods structured through encoder-decoder architectures [19]. The model employs an encoder network to map the input physical states into a latent space. Within this latent domain, linear transformations are applied to the projected variables. Finally, a decoder network maps the latent representation back to the physical space to evaluate the time derivatives of the system.

1. Mapping to the Linear Latent Space
The initial processing of the physical states consists of a structured dimensional projection, specifically designed to avoid the cross-convolution of the gravitational system's degrees of freedom. To project these physical variables into the latent space without compromising the independence of each mass, both branches undergo a point-to-point parametric linear

transformation. Unlike standard dense layers, which couple the positional coordinates of different bodies through matrix multiplication, each degree of freedom in the SLL-HNN is scaled and shifted independently using specific trainable weights and biases. This projection is mathematically defined using Hadamard product operations (element-by-element multiplication), as shown in **Eq. (3)**:

$$\tilde{q} = \hat{q} \odot W_q + b_q, \quad \tilde{p} = \hat{p} \odot W_p + b_p \tag{3}$$

where $W_q, b_q \in \mathbb{R}^{30}$ and $W_p, b_p \in \mathbb{R}^{30}$ are the parametric vectors of weights and biases for positions and momenta, respectively, and $\tilde{q}, \tilde{p} \in \mathbb{R}^{30}$ represent the projected coordinates in the linear latent space. This mechanism guarantees that the positional or inertial influence of each planet is autonomously calibrated in the latent domain before entering the energy estimation modules. The complete topological arrangement of this separation, from the initial linear mapping to the reconstruction of the gradients, is detailed in **Figure 7**.

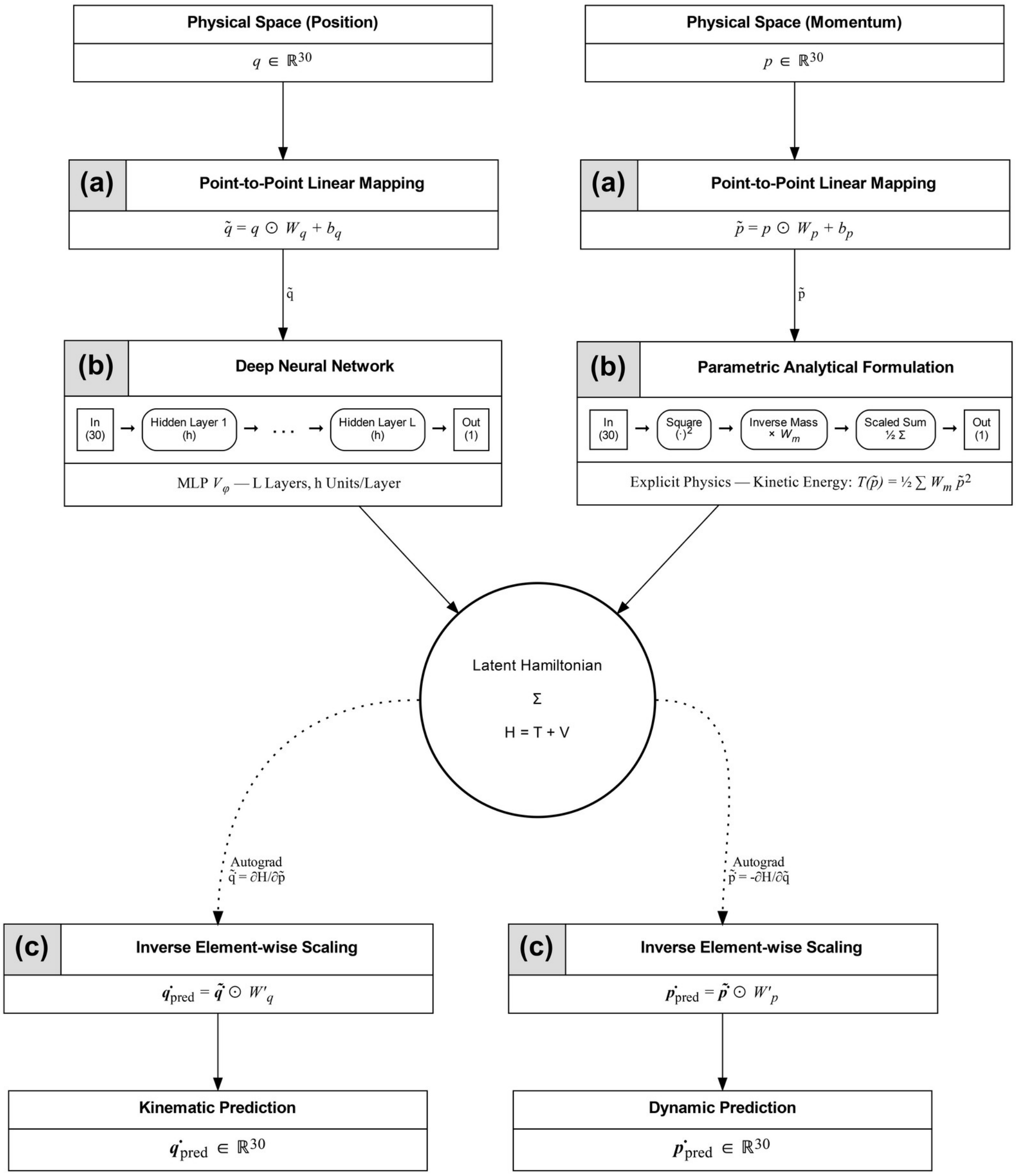


Figure 7: Architecture SLL-HNN. *(a) Independent point-to-point linear projection of position and momentum to the latent space. (b) Decoupled calculation of energies to form the latent Hamiltonian H. (c) Extraction of symplectic gradients and inverse scaling for the final predictions of* $\dot{q}$ *and* $\dot{p}$*.*

2. Internal Configuration of the Subnetworks and synthesis of Latent Hamiltonian

Once the physical states have been projected into the linear latent space, guaranteeing dimensional decoupling, the architecture bifurcates into two specialized submodules. This topological division reflects the additive separability of the Hamiltonian in conservative systems $\left(H=T+V\right)$ as in P-HNN and BLK-P-HNN and allows the application of specific inductive biases for the physical nature of each energy component.

Unlike standard dense layers, which couple the positional coordinates of different bodies through matrix multiplication, each degree of freedom in the SLL-HNN is scaled and shifted independently using specific trainable weights and biases. Consequently, the latent potential energy is parameterized by a standard Multilayer Perceptron (MLP) applied to the latent position vector, where the input and output processing is modified by employing a Hadamard product for this independent linear scaling.

Section 3.5: Global Optimization and Loss Function

Although the operative spaces differ across the proposed architectures utilizing multiscale normalized coordinates $(\hat{q},\hat{p})$ for the P-HNN and BLK-P-HNN, and point-to-point linearly projected variables $(\tilde{q},\ \tilde{p})$ for the SLL-HNN the fundamental learning mechanism remains unified. The networks do not learn the coordinates directly; instead, they optimize the parameterized Hamiltonian surface such that its partial derivatives strictly satisfy Hamilton's canonical equations.

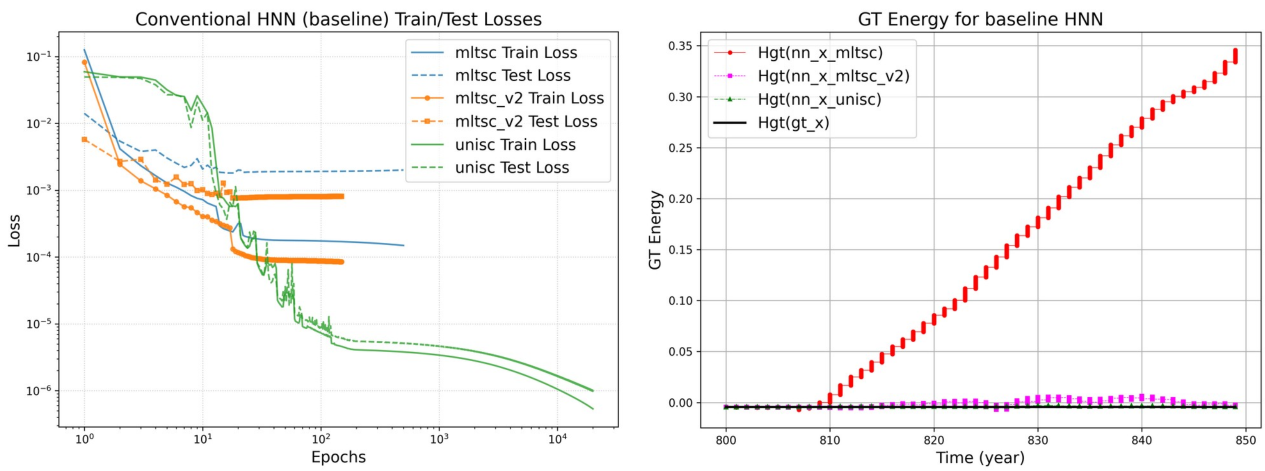
To generalize the optimization objective across all frameworks, let z_q and z_p denote the appropriately transformed position and momentum variables for any given architecture. The total loss function $(\mathcal{L})$ is defined as the Mean Squared Error (MSE) between the network's predicted latent derivatives and the true time derivatives obtained from the numerical solver:

$$\mathcal{L}=\frac{1}{N}\sum_{i=1}^{N}\left(\left\|\dot{z}_{q,i}-\frac{\partial H}{\partial z_{p,i}}\right\|^2+\left\|\dot{z}_{p,i}+\frac{\partial H}{\partial z_{q,i}}\right\|^2\right)$$

This formulation captures both the kinematic evolution and the dynamic force fields. The loss is computed directly without the use of numerical stabilizers (such as ϵ additives) during gradient descent optimization, avoiding the alteration of the physical magnitudes of the 10-body system.

4. Results and Discussions

Section 4.1: Limitations of Baseline HNNs



(a) (b)

Figure 8: Comparison of baseline conventional HNN train/test losses and long-term Hamiltonian energy profiles over the 0–50 year orbit prediction interval across different scaling methodologies. *(a) Evolution of training and testing loss profiles. (b) Ground truth Hamiltonian function computed over true and predicted state variables (q,p). Here, 'unisc' denotes Global Uniscale normalization, 'mltsc' refers to Component-wise per-body scaling, and 'mltsc_v2' represents Vector-wise per-body scaling.*

Initially, we sought to investigate the feasibility of utilizing a conventional Hamiltonian Neural Network (HNN) for the long-term orbital prediction of a multi-body system specifically, a 10-body configuration of the Solar System and to observe its baseline operational characteristics. In their foundational work, Greydanus et al. (2019) configured a network for a 2-body problem by stacking three linear layers with a hidden dimension of 200 neurons per layer. Drawing upon this design rationale and accounting for our 60-dimensional input state vector $(\hat{q}, \hat{p})$ for the 10-body system, we constructed a baseline HNN consisting of three linear layers with 240 hidden neurons per layer to conduct training and evaluation. For this optimization framework, we utilized a 1,000-year synthetic Solar System trajectory dataset, allocating the first 800 years as the training partition and reserving the remaining 200 years exclusively for testing loss measurement and long-term orbit prediction assessment.
To initiate the long-term orbit prediction assessment within the unseen domain, a single data point from the test partition was designated as the initial seed point for the iterative ODE integration. Using this baseline framework, we subsequently analyzed the impact of the three distinct scaling strategies Global Uniscale, Component-wise Multiscale, and Vector-wise Multiscale V2—on the network's optimization path and resulting trajectory stability.
A primary challenge in modeling the Solar System is addressing the significant differences in physical magnitudes across celestial bodies, often characterized as a High Dynamic Range (HDR) problem. To systematically accommodate these varying physical scales, we applied three distinct scaling strategies to both the input state variables $(\hat{q}, \hat{p})$ and the corresponding target time-derivatives $(\dot{\hat{q}}, \dot{\hat{p}})$ across all data samples. These approaches designated as Global Uniscale, Component-wise Multiscale, and Vector-wise Multiscale V2—are formally defined as follows:

$$\begin{aligned}
&\hat{X}_{b,i,j}=\frac{X_{b,i,j}}{s}, \quad \text{with} \quad s=max_{b,i,j}|X_{b,i,j}| \quad \text{(uniscale)}\\
&\hat{X}_{b,i,j}=\frac{X_{b,i,j}}{s_{i,j}}, \quad \text{with} \quad s_{i,j}=max_{b}|X_{b,i,j}| \quad \text{(multiscale)}\\
&\hat{X}_{b,i,j}=\frac{X_{b,i,j}}{s_{i}}, \quad \text{with} \quad s_{i}=max_{b}\sqrt{X_{b,i,x}^2+X_{b,i,y}^2+X_{b,i,z}^2} \quad \text{(multiscale V2)}\\
&\text{where} \quad i\in\{0,1,2,\ldots,9\}, \quad j\in\{x,y,z\}, \quad \text{and} \quad b=sample\,ID
\end{aligned} \tag{4}$$

To evaluate the efficacy of conventional Hamiltonian Neural Networks (HNNs) under highly disparate physical scales, we analyzed the learning convergence, corresponding trajectory profiles, and long-term energy behaviors across the three normalization strategies (**Figure 8** , **Figure 9**, and **Figure 10**).

When employing Global Uniscale normalization, the system faces severe optimization stagnation due to the unmitigated High Dynamic Range (HDR) characteristics. To prevent immediate gradient explosion, an extremely conservative learning rate sequence (decaying to a minimum boundary of 1×10^{-6}) with a large batch size of 8,192 was employed. Under this constrained optimization layout, the baseline architecture demands approximately 20,000 epochs to establish a parameter space capable of maintaining a structurally bound state for a 50-year interval. However, extended training beyond this window yields no discernible performance improvements, identifying a practical orbit prediction boundary at roughly 50 years before numerical precession induces systemic disintegration.

Conversely, both the Component-wise Multiscale and Vector-wise Multiscale V2 frameworks resolve this inherent gradient stiffness by providing improved numerical conditioning across the feature coordinates. This mitigation forces an accelerated optimization path, triggering a steep drop in both training and testing loss bounds within fewer than 200 epochs. Yet, this rapid loss reduction fails to translate into physical validity over extended temporal horizons, revealing distinct modes of energetic failure. As mapped in the long-term energy tracking (**Figure 9**), the standard Component-wise Multiscale model undergoes a continuous, linear escalation in total Hamiltonian energy after a 10-year integration threshold, causing immediate structural decay. In contrast, the Vector-wise Multiscale V2 architecture yields bounded energy deviations that oscillate near the true Hamiltonian value; however, these geometric fluctuations still introduce localized instabilities over time. Ultimately, both scaling mechanics lead to a cascading destabilization event, culminating in a domino effect of unphysical inner planetary ejections within the localized domain (**Figure 10**).

Based on these findings, the Vector-wise Multiscale V2 was selected as our foundational normalization standard due to its superior learning convergence and relative energy control properties. This framework was subsequently applied to both the Plausible HNN (as shown conceptually in **Figure 4 (b)**) and Block Plausible HNN architectures (**Figure 5** and **Figure 6**) to structurally resolve the remaining trajectory drifts. To fully exploit these scaling characteristics, we first introduced the Plausible HNN (P-HNN), shifting away from generic black-box networks by enforcing an explicit Kinetic-Potential (T-V) energy separation and functional symmetry.

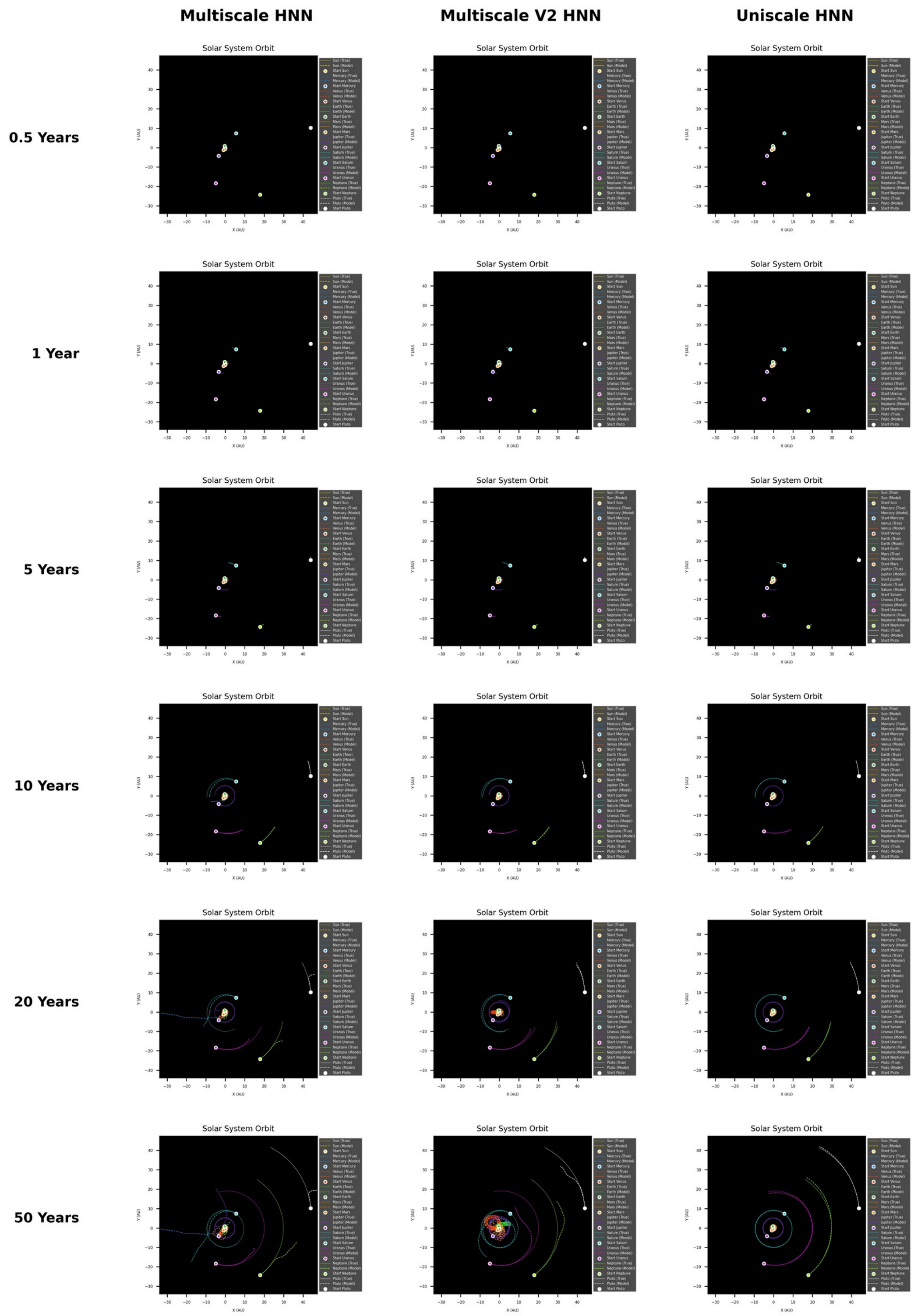


Figure 9: 50 years of orbit predictions for the Solar System with baseline HNNs. Macro-view of 10-body of the Solar system orbit predictions by feeding HNNs with the 1st point of test region dataset. *Rows denote integration spans (0.5 to 50 years), and columns represent scaling formulations within the baseline dense HNN. The models were trained using the first 800 years data and evaluated for the remaining 200 years data from 1000 years dataset in Δt=0.0005.*

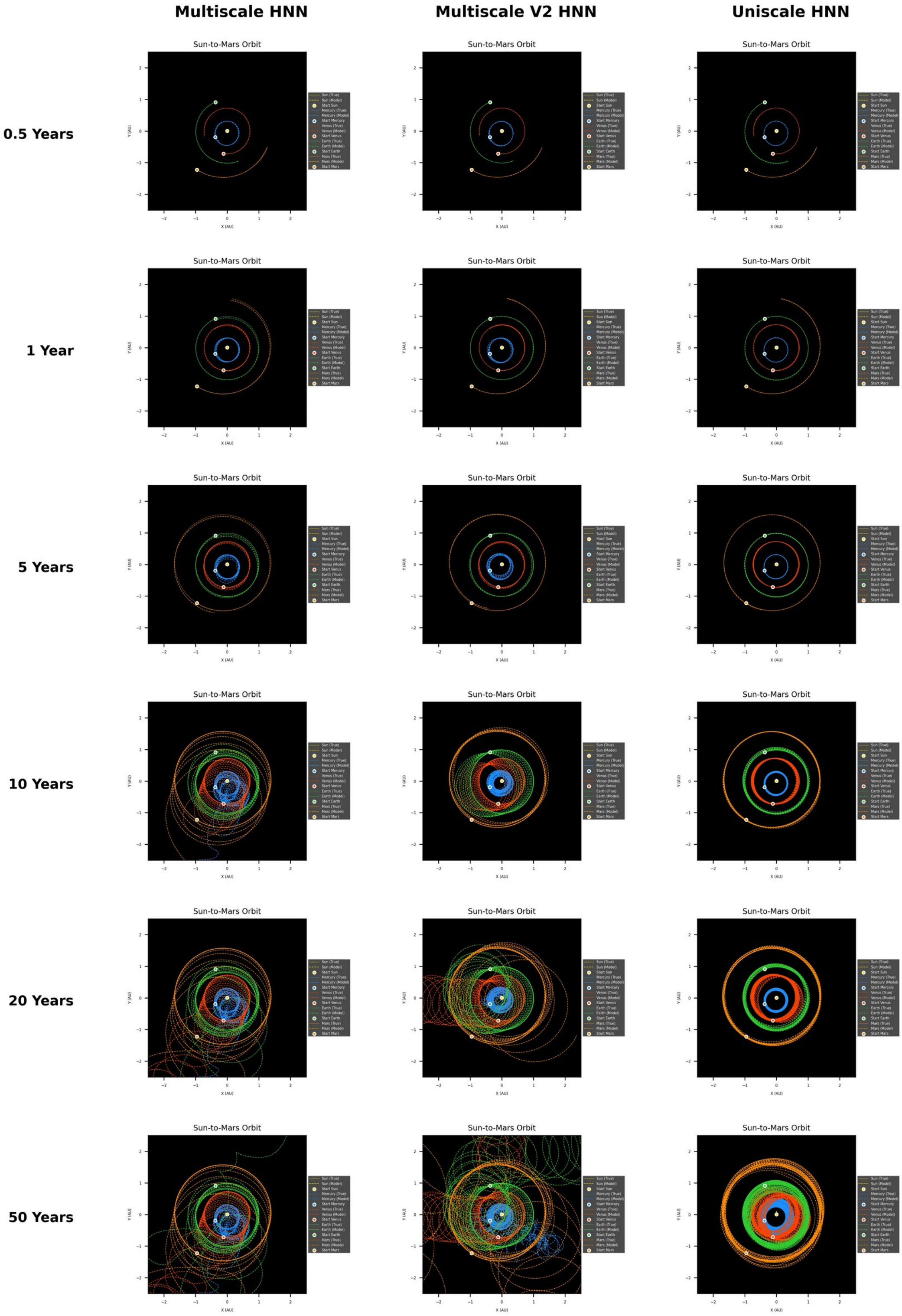


Figure 10: Predicted inner-planetary orbits detailed, highlighting rapid structural decay. Close-up view focusing on high-frequency inner planets (Mercury to Mars) under identical time horizons and scaling strategies as Figure 4. *This demonstrates instantaneous trajectory ejections in 5 to 10 years for rapid-converging multiscale models. The models were trained using the first 800 years data and evaluated for the remaining 200 years data from 1000 years dataset in Δt=0.0005.*

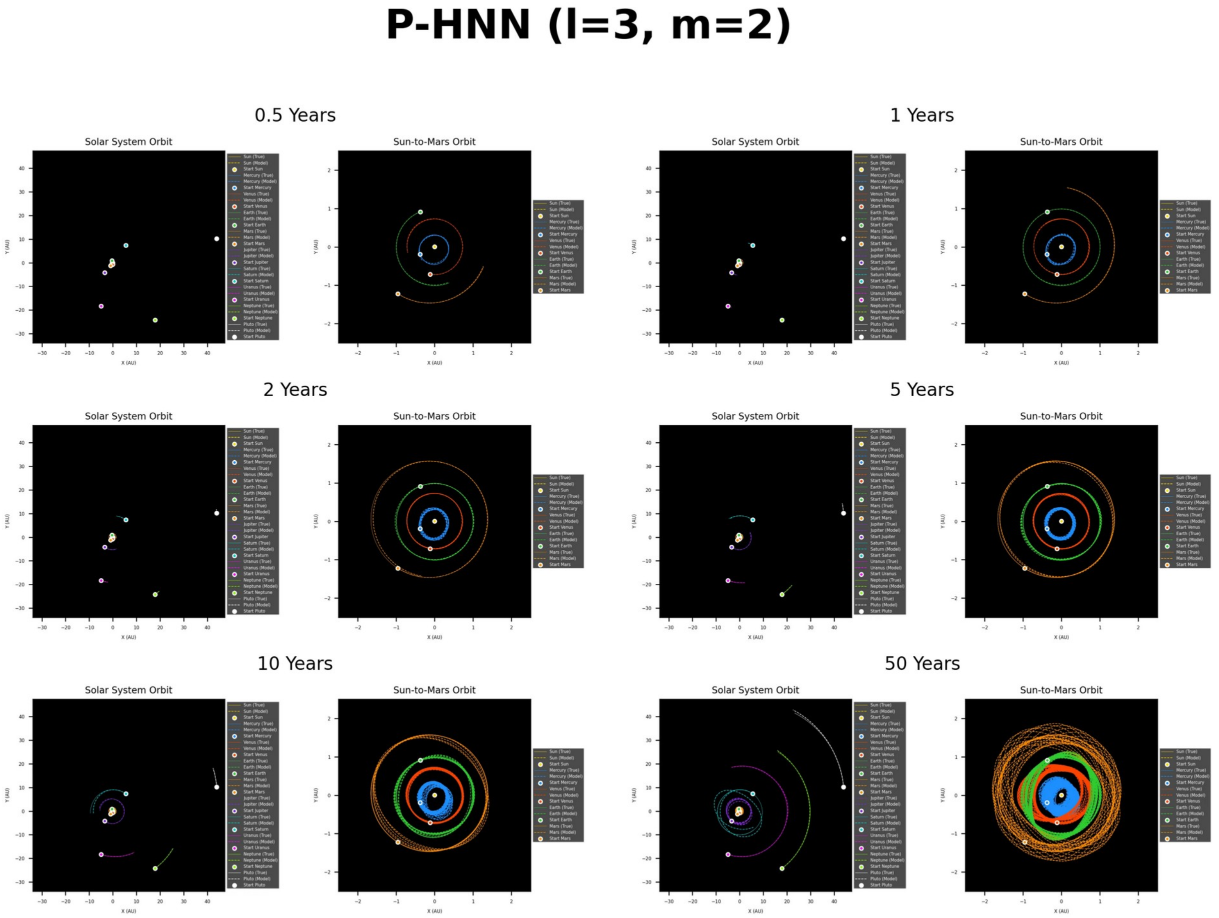


Figure 11: 50 years of orbit prediction by PlausibleHNN (P-HNN) for layers l=3 and factor m=2, which corresponds to 270 hiddens per layer. *The model was trained using the first 800 years data and evaluated for the remaining 200 years data from 1000 years dataset in Δt=0.0005.*

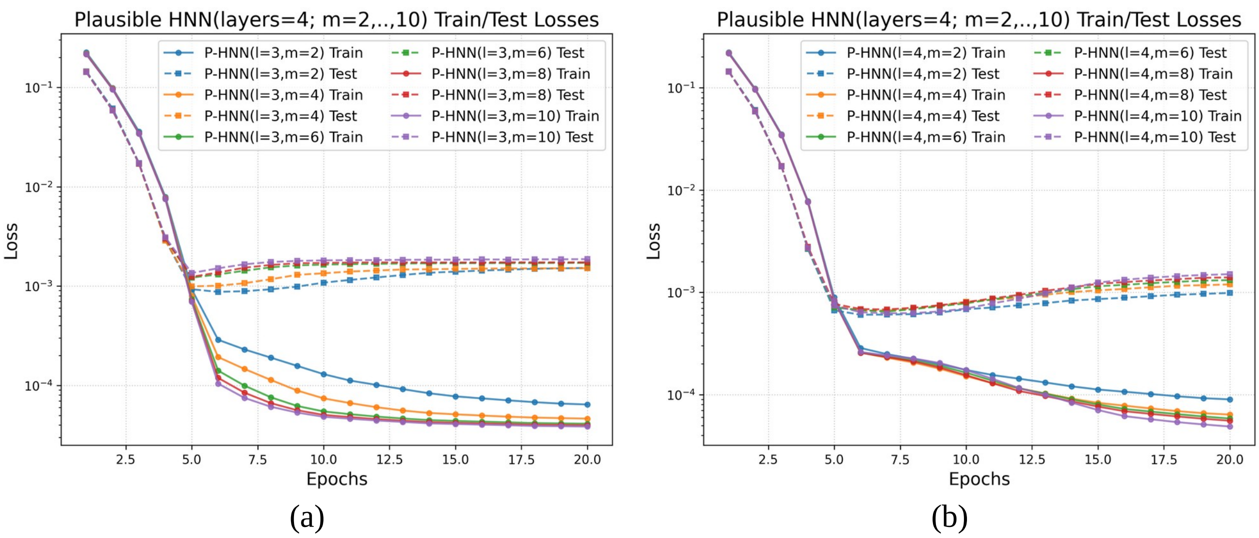


Figure 12: PlausibleHNN train/test losses: (a) layers l=3, factor m=2, 4,..., 10 and (b) layers l=4, factor m=2, 4, ..., 10.

4.2 Performance Analysis of the Plausible HNN (P-HNN)

To establish a controlled evaluation, the initial Plausible HNN (P-HNN) configuration was designed to match the baseline dense model by employing three hidden layers with a scaling factor of $m=2$ ($l=3, m=2$), which corresponds to a hidden dimension of 270 units (${}_{10}C_2 \times 3 \times 2 = 270$). Under the Vector-wise Multiscale V2 scheme, this initial baseline P-HNN successfully maintains a gravitationally bound state for approximately 50 years (**Figure 11**), replicating the maximum lifespan achieved by the slow-converging Uniscale baseline but with an accelerated optimization path (< 20 epochs; **Figure 12**).

To improve the model's trajectory representation, we systematically expanded the network capacity by scaling the layer width parameter across $m \in \{4, 6, 8, 10\}$. While increasing the internal features generally showed a tendency to extend the orbital lifespan, the $l=3$ dense configuration encountered a strict operational ceiling and failed to sustain stable bounds beyond the 200-year horizon except for the $l=3$ and $m=8$ case (**Figure 13** and **Figure 14**).

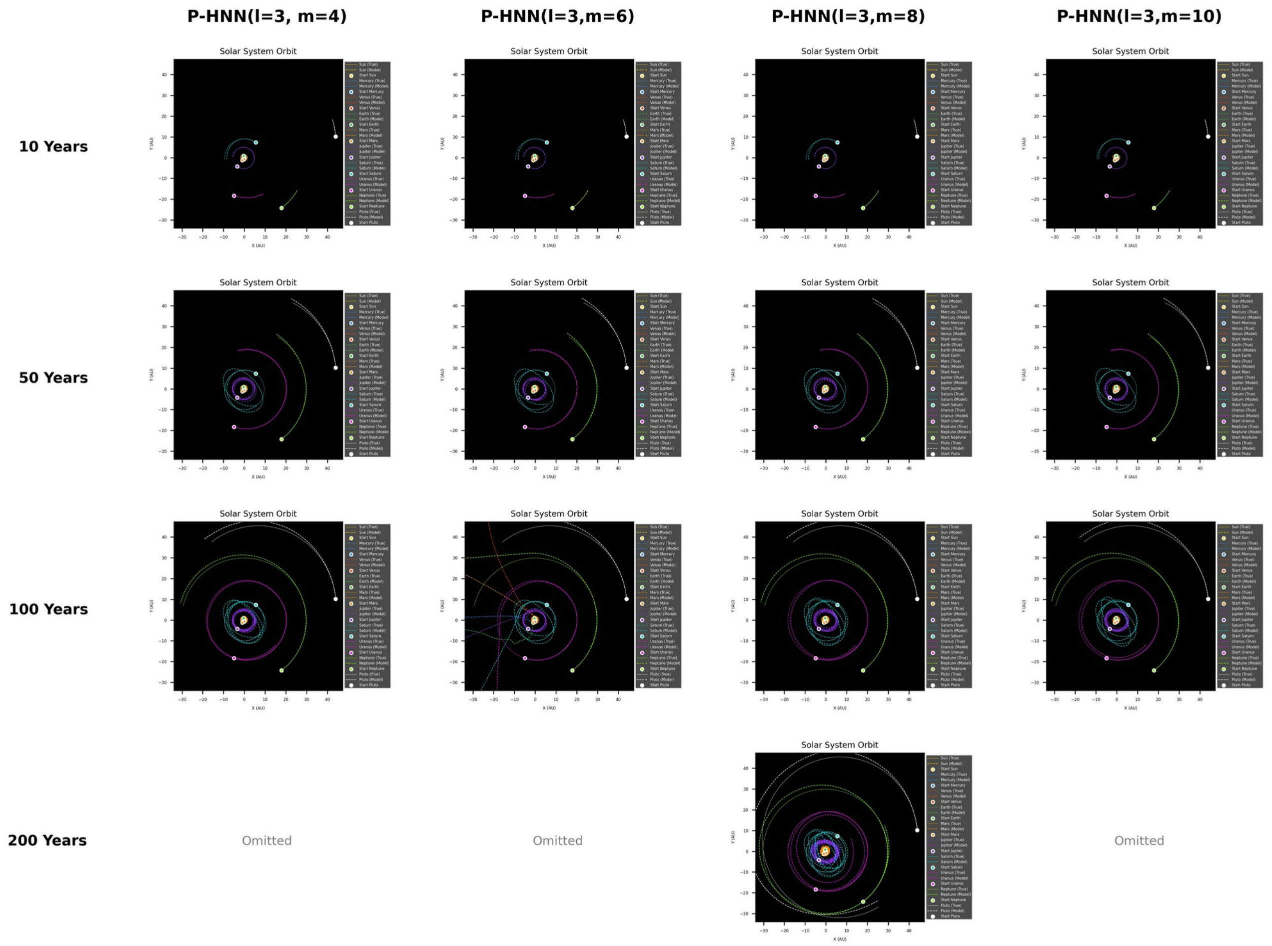


Figure 13: 200 years of orbit predictions for the Solar System with PlausibleHNN (P-HNN) for layers l=3 and factor m=4, 6, 8, 10 by feeding P-HNNs with the 1st point of test region dataset. *Apparently disintegrated orbits are not shown (omitted). The models were trained using the first 800 years data and evaluated for the remaining 200 years data from 1000 years dataset in Δt=0.0005.*

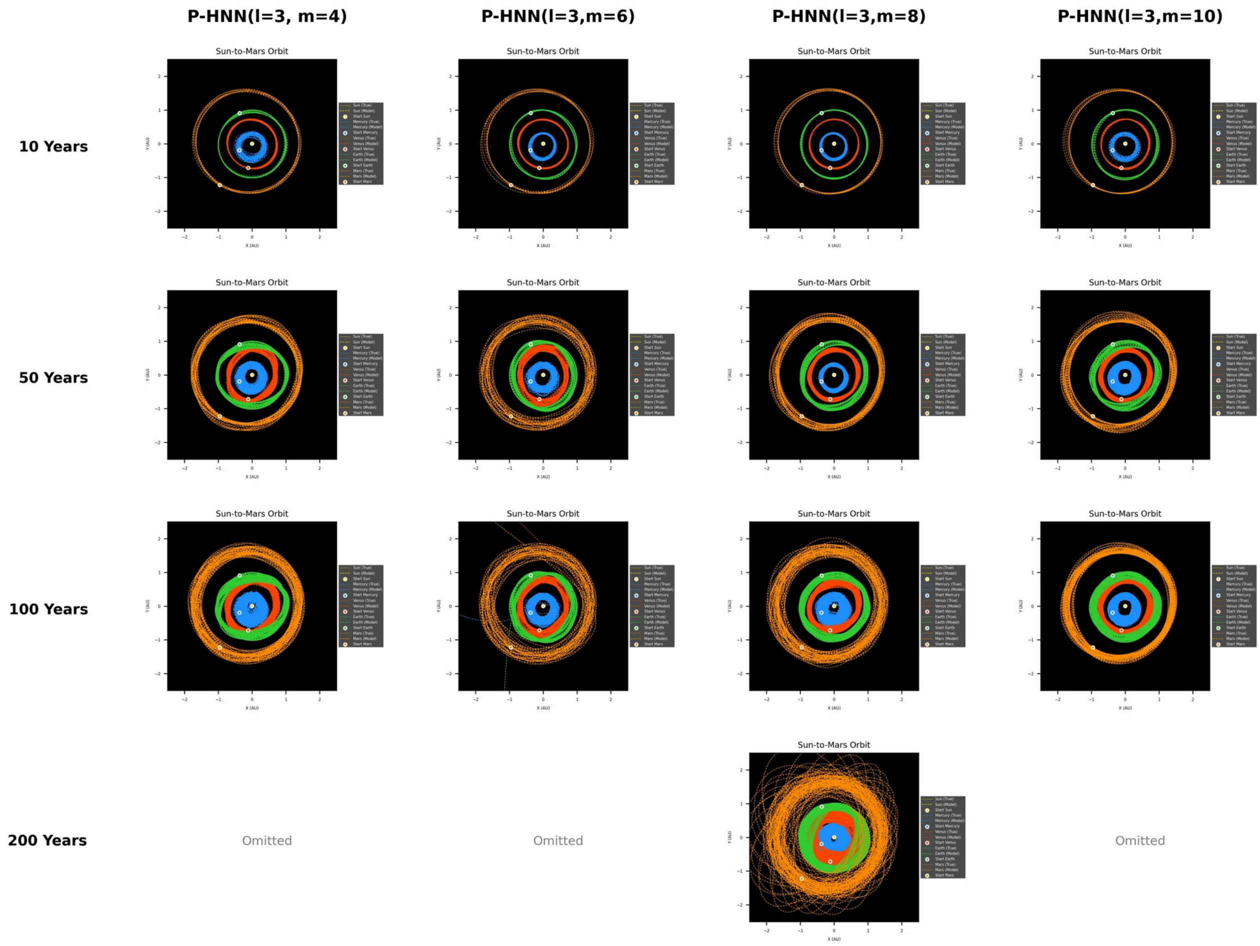


Figure 14: Detailed inner-planetary orbits predicted for 200 years with PlausibleHNN (P-HNN) for layers l=3 and factor m=4, 6, 8, 10 by feeding P-HNNs with the 1st point of test region dataset. *Apparently disintegrated orbits are not shown (omitted). The models were trained using the first 800 years data and evaluated for the remaining 200 years data from 1000 years dataset in Δt=0.0005.*

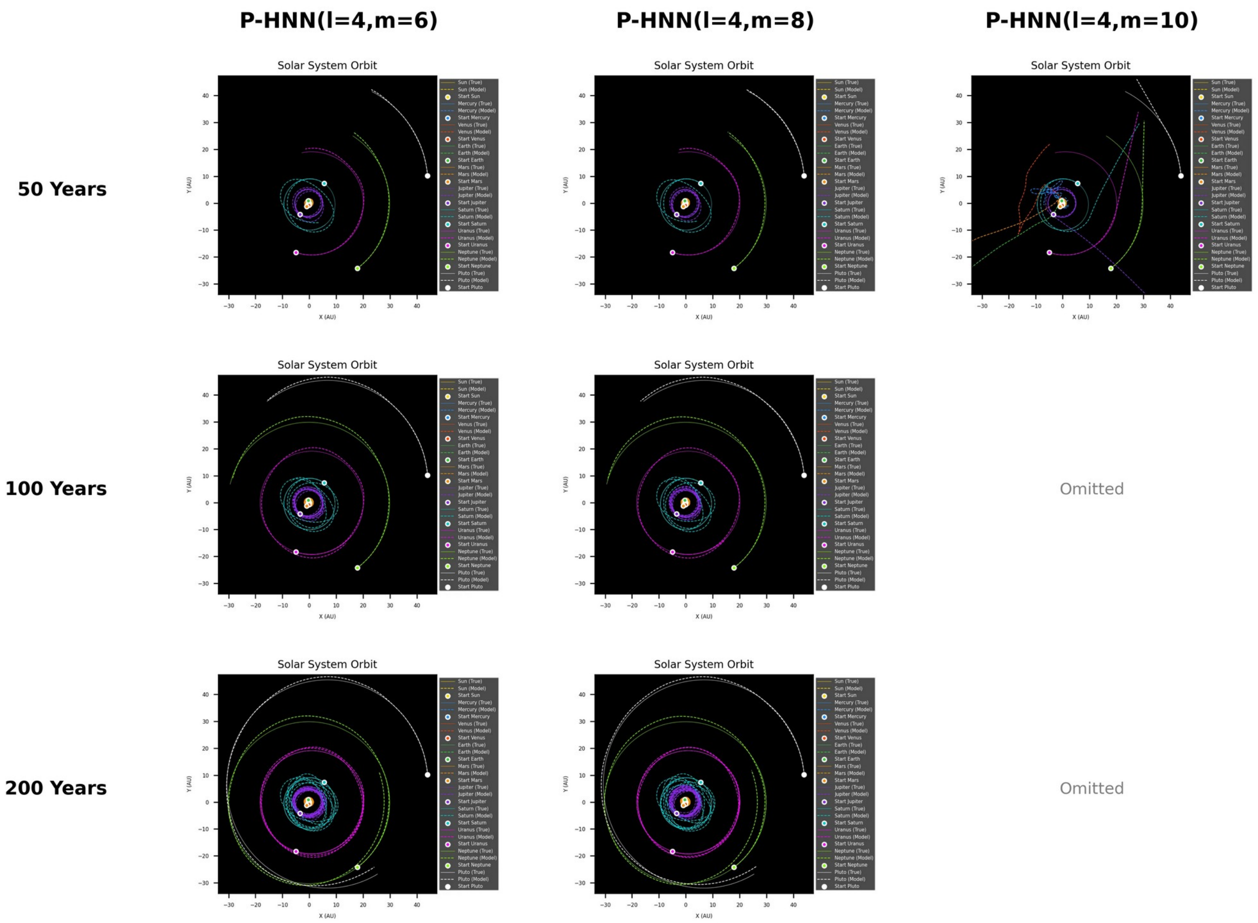


Figure 15: 200 years of orbit predictions for the Solar System with PlausibleHNN (P-HNN) for layers l=4 and factor m=6, 8, 10 by feeding P-HNNs with the 1st point of test region dataset. *Apparently disintegrated orbits are not shown (omitted). The models were trained using the first 800 years data and evaluated for the remaining 200 years data from 1000 years dataset in Δt=0.0005.*

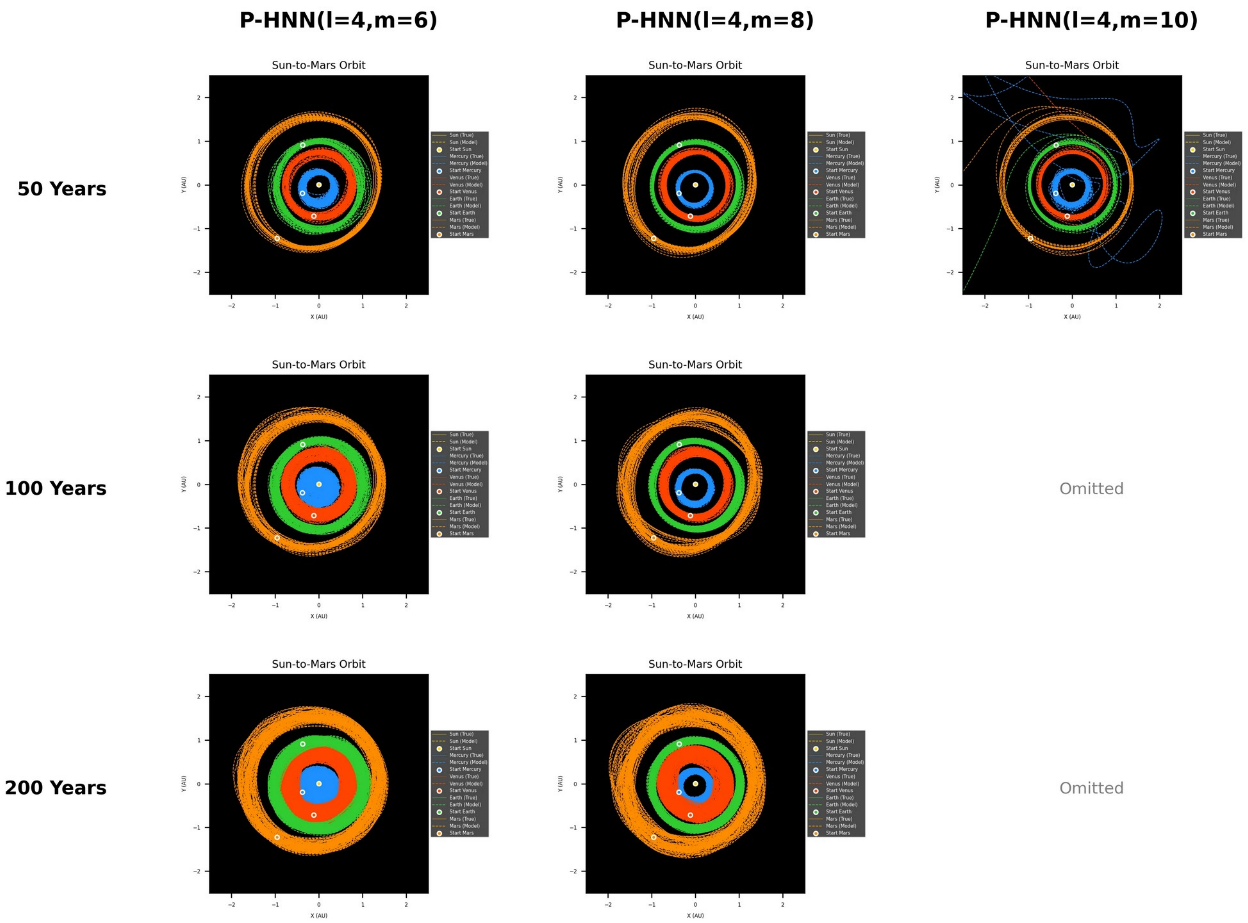


Figure 16: Detailed inner-planetary orbits predicted for 200 years with PlausibleHNN (P-HNN) for layers l=4 and factor m=6, 8, 10 by feeding P-HNNs with the 1st point of test region dataset. *Apparently disintegrated orbits are not shown (omitted). The models were trained using the first 800 years data and evaluated for the remaining 200 years data from 1000 years dataset in Δt=0.0005.*

To break this geometric barrier, the structural depth was increased by adding an extra hidden layer ($l=4$). Empirically, the four-layer configuration with an enhanced feature width ($l=4, m=8$) successfully extended the system-level lifespan up to 200 years within the unseen test domain (**Figure 15** and **Figure 16**). Throughout this 200-year window, the planets experienced gradual, bounded numerical precession without catastrophic ejections. This explicit capacity expansion to ($l=4, m=8$), however, magnified the model's parametric footprint to approximately 57 MB.

The 200-year lifespan achievement of the dense P-HNN ($l=4, m=8$) of this 57 MB model size, represents a substantial advancement over the baseline configurations. However, it still falls short of capturing a complete orbital cycle of the farthest member in our 10-body setup: Pluto, which possesses an orbital period of approximately 248 years. For a learned Hamiltonian system to be considered dynamically complete and structurally stable over astronomical timescales, the integration window must surpass this 248-year threshold to ensure that the slowest high-dimensional phase space couplings are fully generalized rather than temporarily fitted.

To cover the prediction for at least one complete revolution of Pluto, we extended the dataset timeline up to 4,000 years using a 12th-order Yoshida integrator, allocating the first 3,200 years for training and the remaining 800 years for testing and evaluation. However, processing this dense model over a high-fidelity 4,000-year dataset generated with a step size of Δt=0.0005 (2,000 steps per year) created an intractable computational overhead, as the raw files reached a massive 6 GB in total size (with $q, p, \dot{q}, \dot{p}$ occupying 1.5 GB each).

To resolve this overhead and facilitate rapid iterative validation, the 4,000-year temporal data was downsized by a factor of 20, sampling every 20th step to achieve a temporal resolution of Δt=0.01 (100 steps per year). This downsampling effectively compressed the dataset footprint to approximately 402 MB (with $q, p, \dot{q}$, and $\dot{p}$ reduced to roughly 96 MB each). Crucially, verifying the dense P-HNN ($l=4, m=8$) architecture on this downsized temporal resolution proved that the underlying physics remained invariant. The model successfully validated extended tracking capabilities, demonstrating that a specific deep configuration could sustain the bound state up to 400 years (**Figure 17** through **Figure 20**).

When the P-HNN models were trained with 3200 years and evaluated the remaining 800 years using the downsized 4000 years dataset with Δt=0.01, we got P-HNN$(l=4, m=10)$ of 88.6MB model size, as the best model this time, in view of overall orbit prediction performances in subsequent test partitions of 400 years window each: test partition 1 ranging from 3200 to 3600 years and the test partition 2 ranging from 3600 to 4000 years.

As illustrated in **Figure 21**, the energy profiles computed via both the ground truth Hamiltonian and the learned P-HNN Hamiltonian remain conserved with only minute fluctuations. This tight energetic alignment indicates that the overall multi-body system is successfully stabilized, preventing long-term structural collapse or unphysical planetary disintegration.

Nonetheless, the massive 88.7 MB memory footprint of this dense layout during extended inference motivated the final structural shift toward sparse partitioning. To surpass this boundary while simultaneously lifting the massive 88.7 MB computational load of the dense layout, we introduced the Block (Sparse) Plausible HNN (BLK-P-HNN) in the subsequent stage.

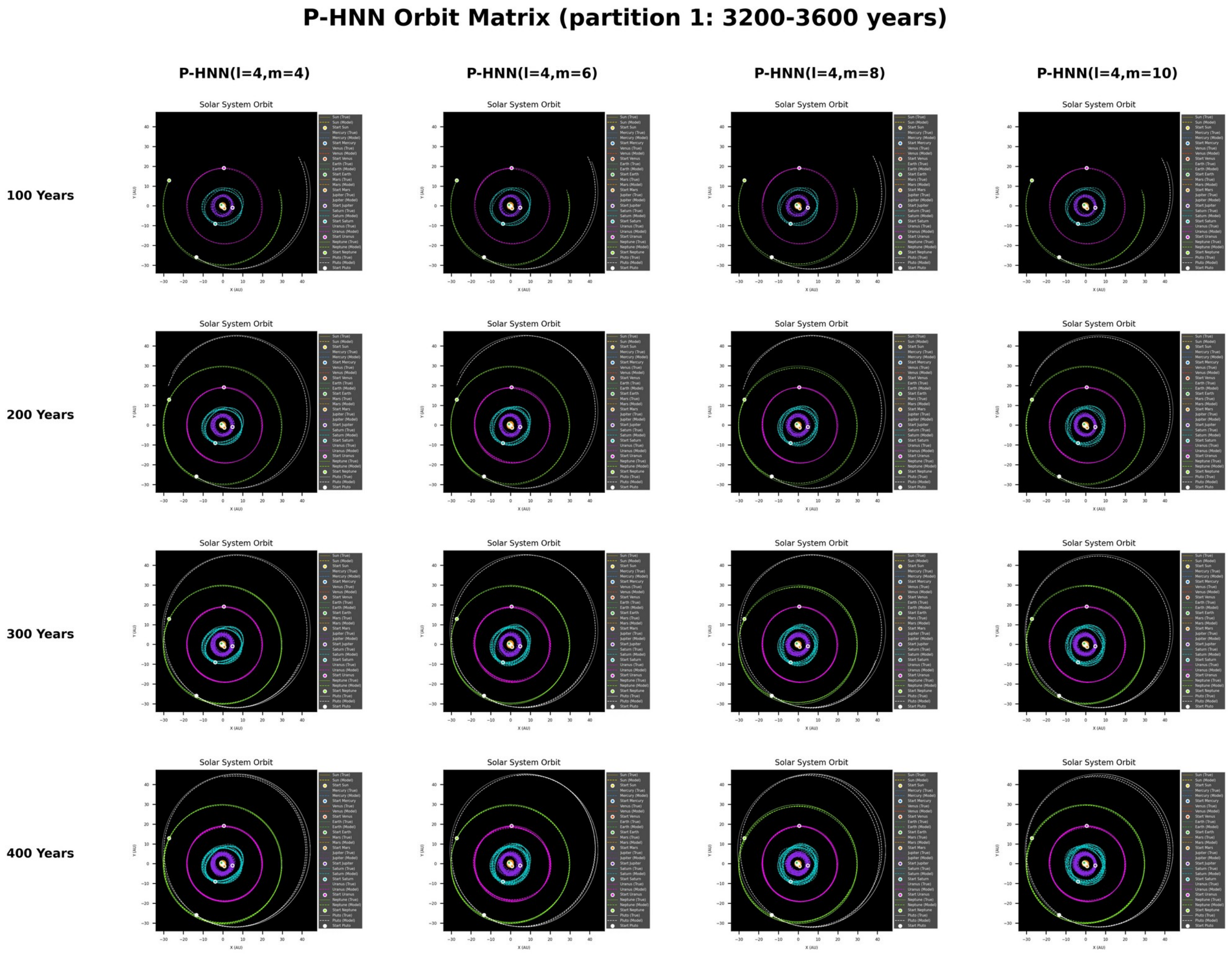


Figure 17: 400 years of orbit predictions for the Solar System with PlausibleHNN (P-HNN) for layers l=4 and factor m=4, 6, 8, 10 by feeding P-HNNs with the 1st point of test partition 1 ranging from 3200 to 3600 years in downsized dataset with Δt=0.01. *The models were trained using the first 3200 years data and evaluated for the remaining 800 years data from 4000 years dataset in Δt=0.01.*

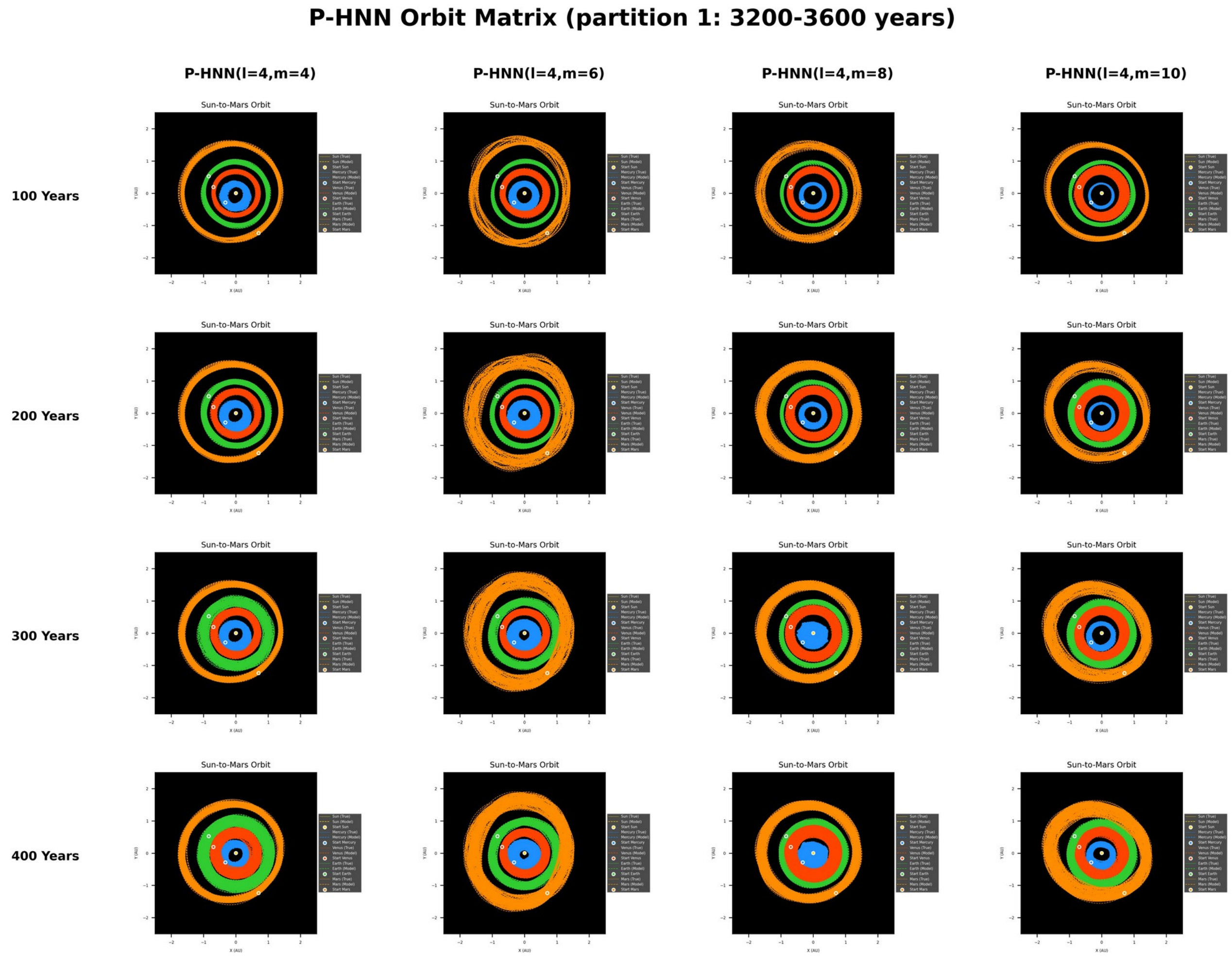


Figure 18: Detailed inner-planetary orbits predicted for 400 years with PlausibleHNN (P-HNN) for layers l=4 and factor m=4, 6, 8, 10 by feeding P-HNNs with the 1st point of test partition 1 ranging from 3200 to 3600 years. *The models were trained using the first 3200 years data and evaluated for the remaining 800 years data from 4000 years dataset in Δt=0.01.*

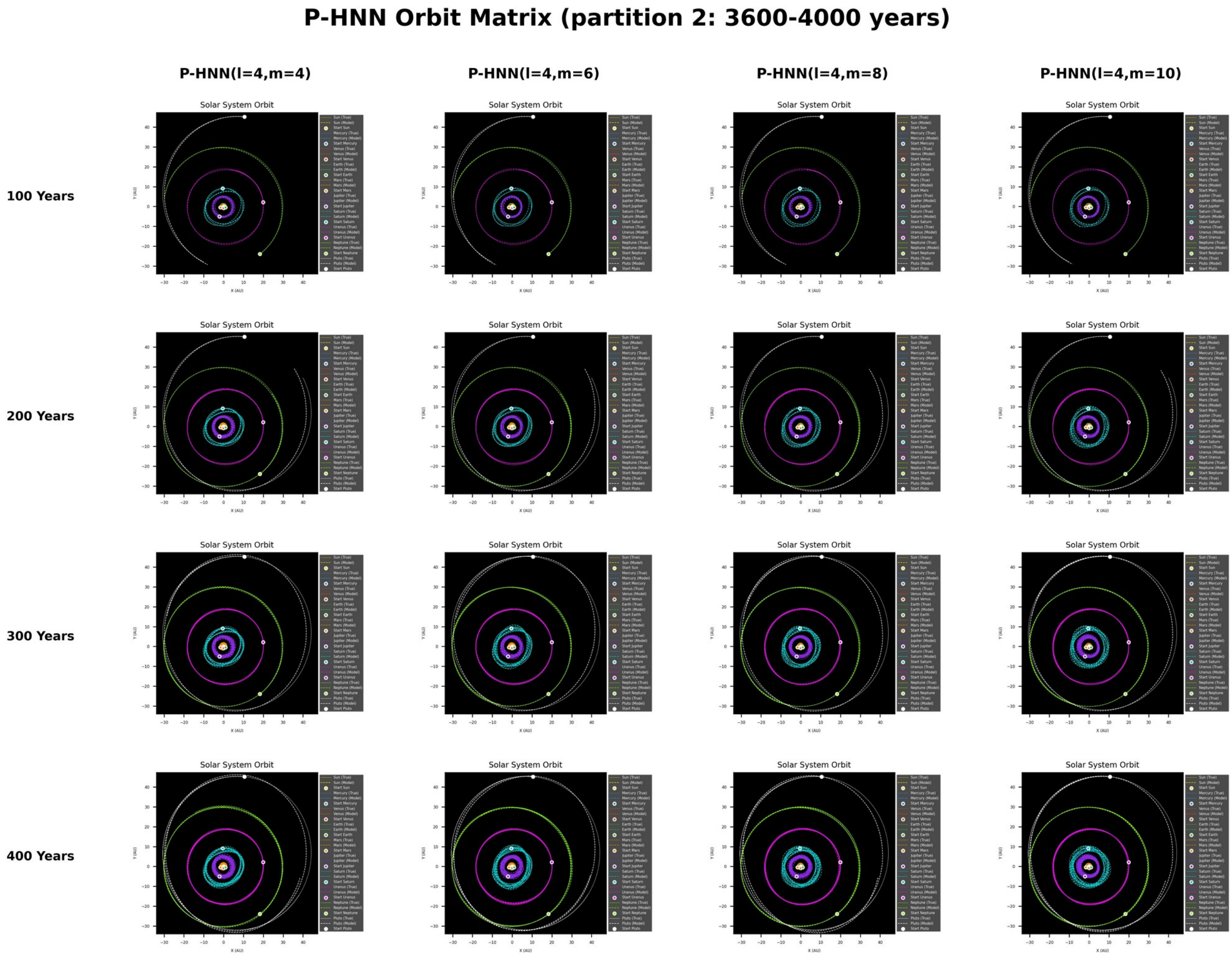


Figure 19: 400 years of orbit predictions for the Solar System with PlausibleHNN (P-HNN) for layers l=4 and factor m=4, 6, 8, 10 by feeding P-HNNs with the 1st point of test partition 2 ranging from 3600 to 4000 years in downsized dataset with Δt=0.01.*The models were trained using the first 3200 years data and evaluated for the remaining 800 years data from 4000 years dataset in Δt=0.01.*

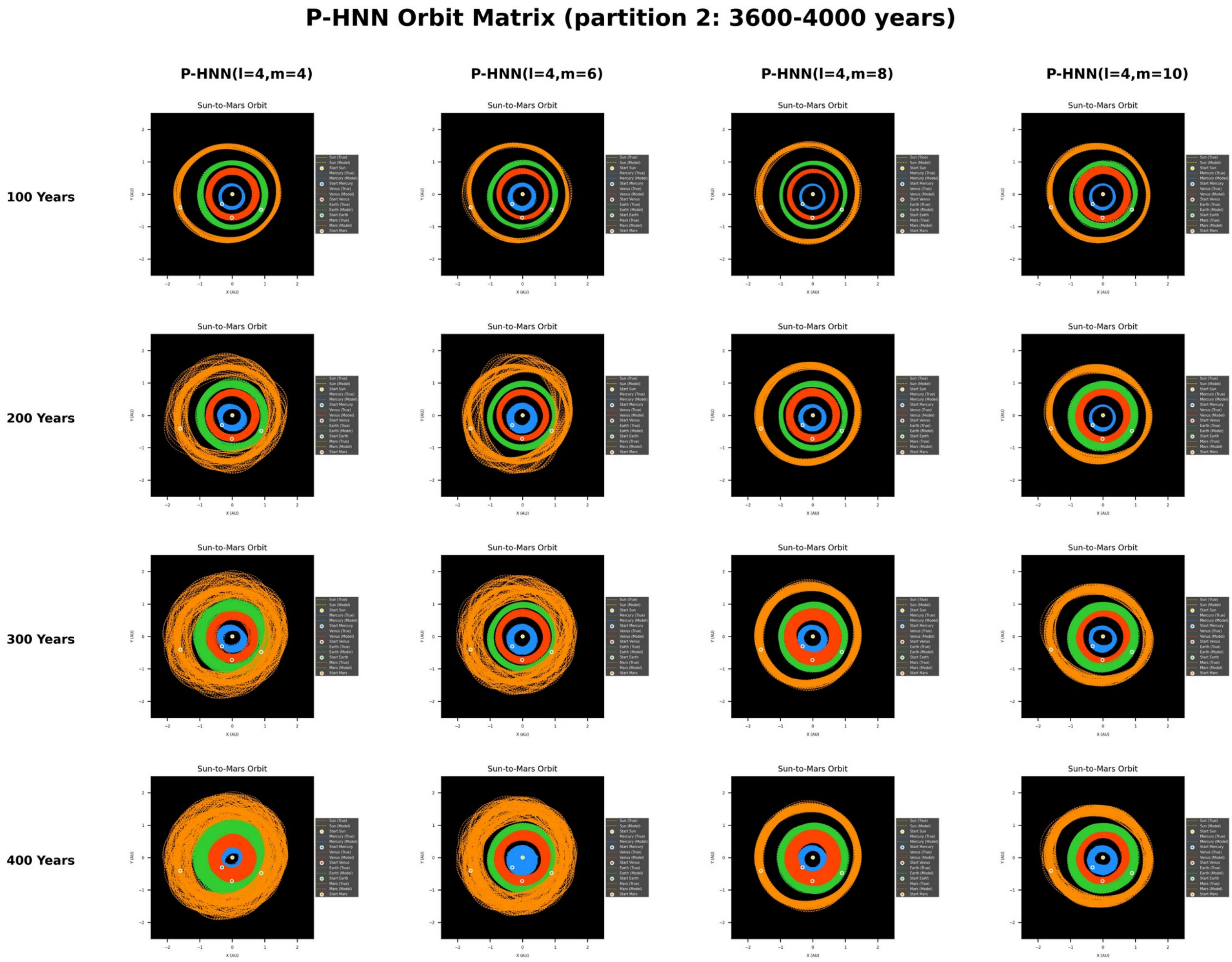


Figure 20: Detailed inner-planetary orbits predicted for 400 years with PlausibleHNN (P-HNN) for layers l=4 and factor m=4, 6, 8, 10 by feeding P-HNNs with the 1st point of test partition 2 ranging from 3600 to 4000 years. *The models were trained using the first 3200 years data and evaluated for the remaining 800 years data from 4000 years dataset in Δt=0.01.*

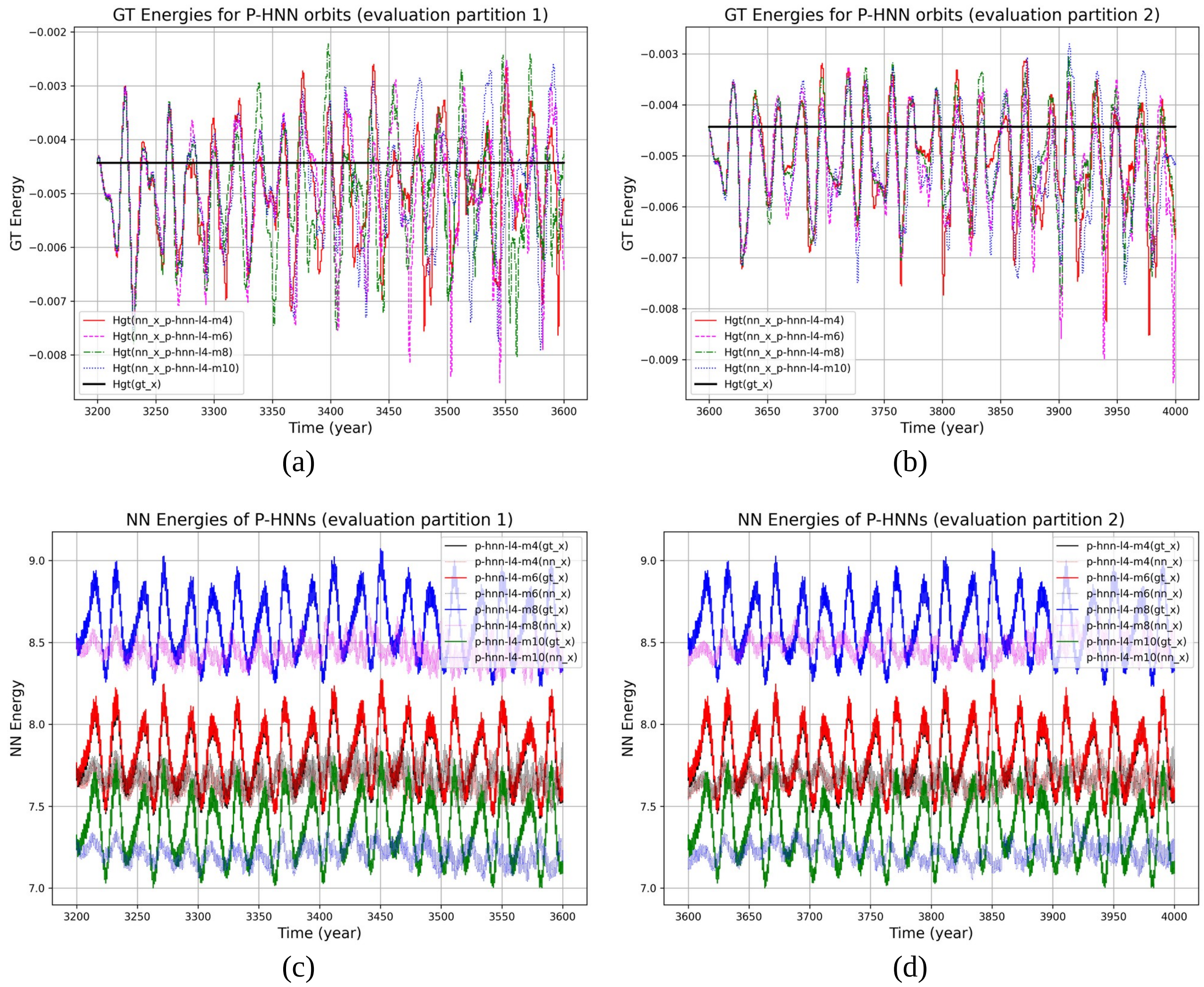
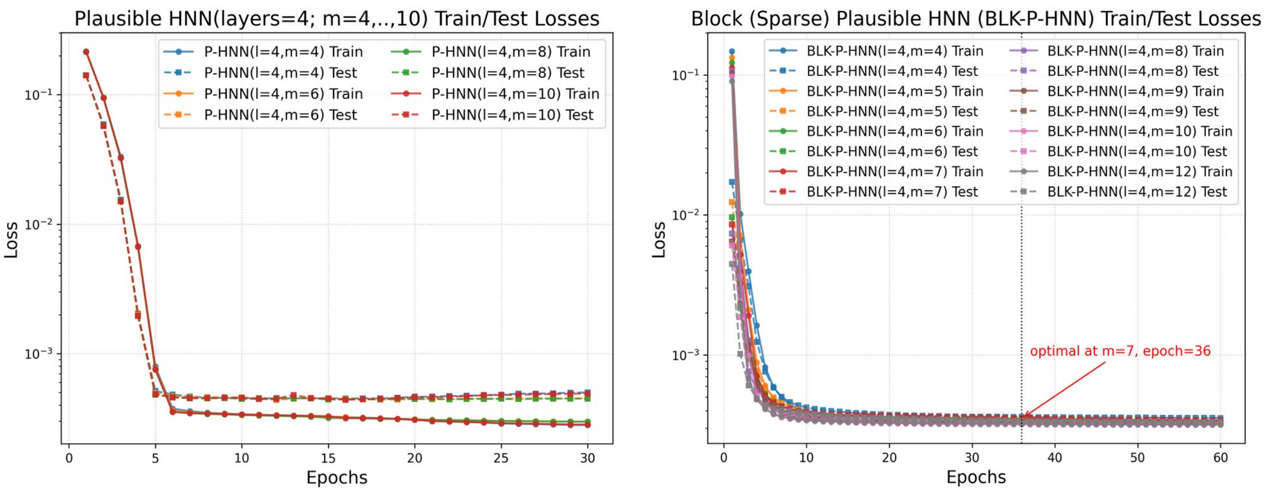


***Figure 21: True and Network energies for true orbits and orbits predicted by PlausibleHNN (P-HNN) of layers l=4 and m=4, .., 10**: (a) True energies for 3200-3600 years, (b) true energies for 3600-4000 years, (c) network energies for 3200-3600 years, and (d) network energies for 3600-4000 years.*

4.3 Structural Optimization via Block (Sparse) Plausible HNN (BLK-P-HNN)

To address the computational bottleneck and memory footprint of the dense P-HNN without degrading its 400-year geometric stability, we evaluated the Block (Sparse) Plausible HNN (BLK-P-HNN) framework as shown in **Figure 5** and **Figure 6**. We considers only $l=4$ cases which have exhibited better performances in the Plausible HNNs than $l=3$ cases.

Interestingly, the train/test loss profiles of P-HNN and BLK-P-HNN differ on the downsized 4000 years dataset as illustrated in **Figure 22**. In case of P-HNN, the best model for each *l* and *m* can be easily found by taking it at the epoch where the minimum of the test loss occurs. On the other hand, even after a sufficient initial decrease, the train/test losses of BLK-P-HNN tend to continuously decline in a counterintuitive manner as training progresses. However, when inspecting the actual orbit prediction trajectories using weights from each subsequent epoch, a lower test loss did not necessarily translate to a superior model. Instead, the optimal model for each l and m within the BLK-P-HNN framework was found near the midpoint of the inspection window. This phenomenon indicates that the optimization landscape of BLK-P-HNN is implicitly ill-conditioned, suggesting that the optimal solution must be sought by intentionally truncating the iterations. This behavior closely resembles the L-curve criterion widely utilized in standard ill-posed inverse problems to determine the optimal regularization cutoff via iteration truncation [14, 15]. In this manner, the optimal model weight for BLK-P-HNN was obtained at $m=7$, in the aspect of the overall prediction performances throughout the evaluation partition 1 ranging from 3200 to 3600 years and the subsequently evaluation partition 2 ranging from 3600 to 4000 years. The orbit predicted with BLK-P-HNN ($l=4, m=7$) for 400 years, energies, and errors in positions/momenta is illustrated in **Figure 23** and **Figure 24**.



(a) (b)

Figure 22: Train/test losses on the downsized 4000 years dataset with Δt=0.01 for (a) Plausible HNN (P-HNN) and (b) Block Sparse Plausible HNN (BLK-P-HNN).

By utilizing the *Topology Adaptor,* which works as a pair-wise input linear adaptor, to isolate the position inputs into 45 independent sub-manifolds via batch matrix multiplications, the BLK-P-HNN effectively eliminates redundant cross-planet weight connections. This structural sparsity compresses the best model's total weight footprint from 88.6 MB at P-HNN($l=4, m=10$) down to 200 kB at BLK-P-HNN($l=4, m=7$)**,** representing an approximate 400-fold parameter reduction. When subjected to the 400-year integration test window (partition 1 and 2 ranging from 3200 to 3600 years and from 3600 to 4000 years respectively), the sparse model still shows excellent generalization capabilities.

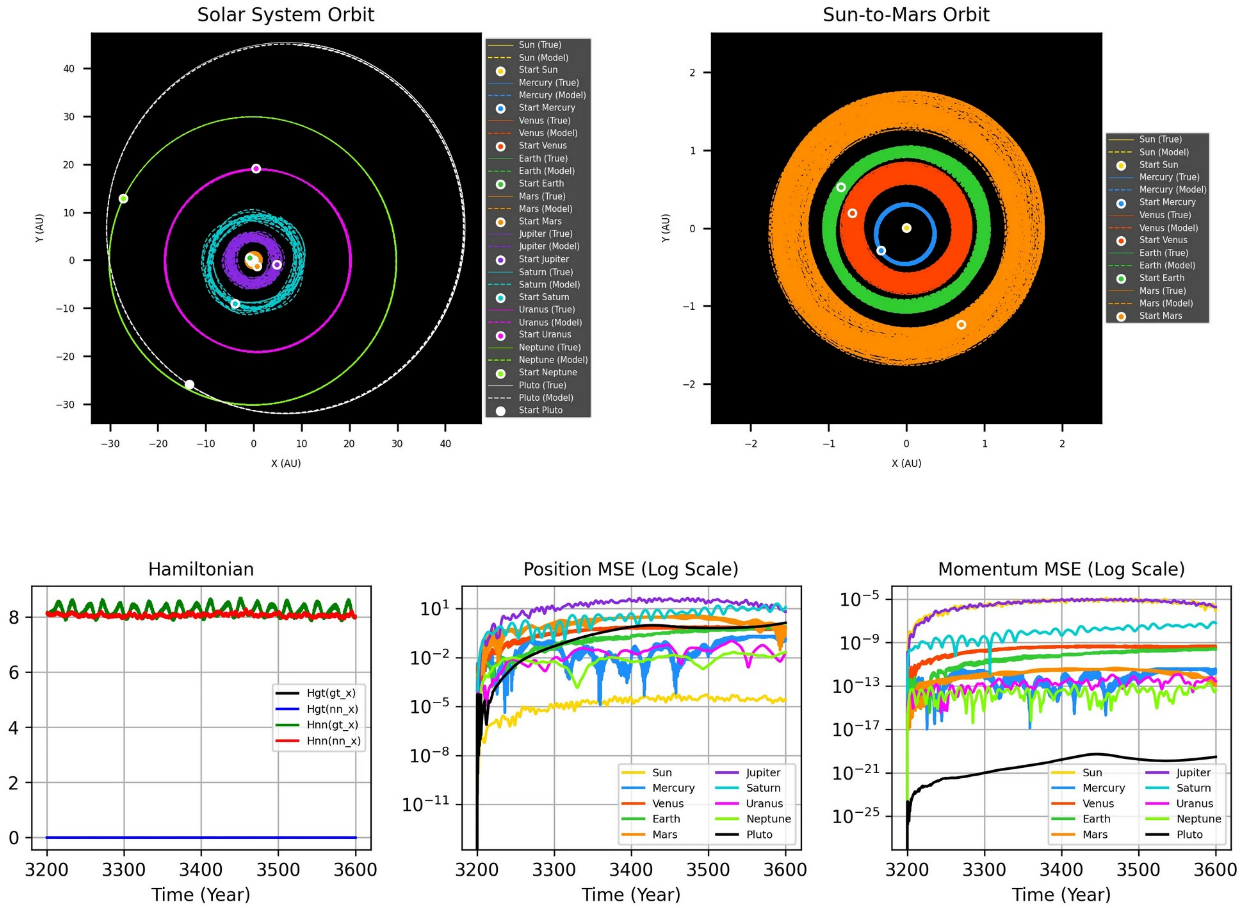


Figure 23: 400 years of orbit predictions and metrics for Block (Sparse) Plausible HNN (BLK-P-HNN) at the optimal configuration (layers l=4 and factor m=7) at evaluation partition 1 ranging from 3200 to 3600 years with Δt=0.01.

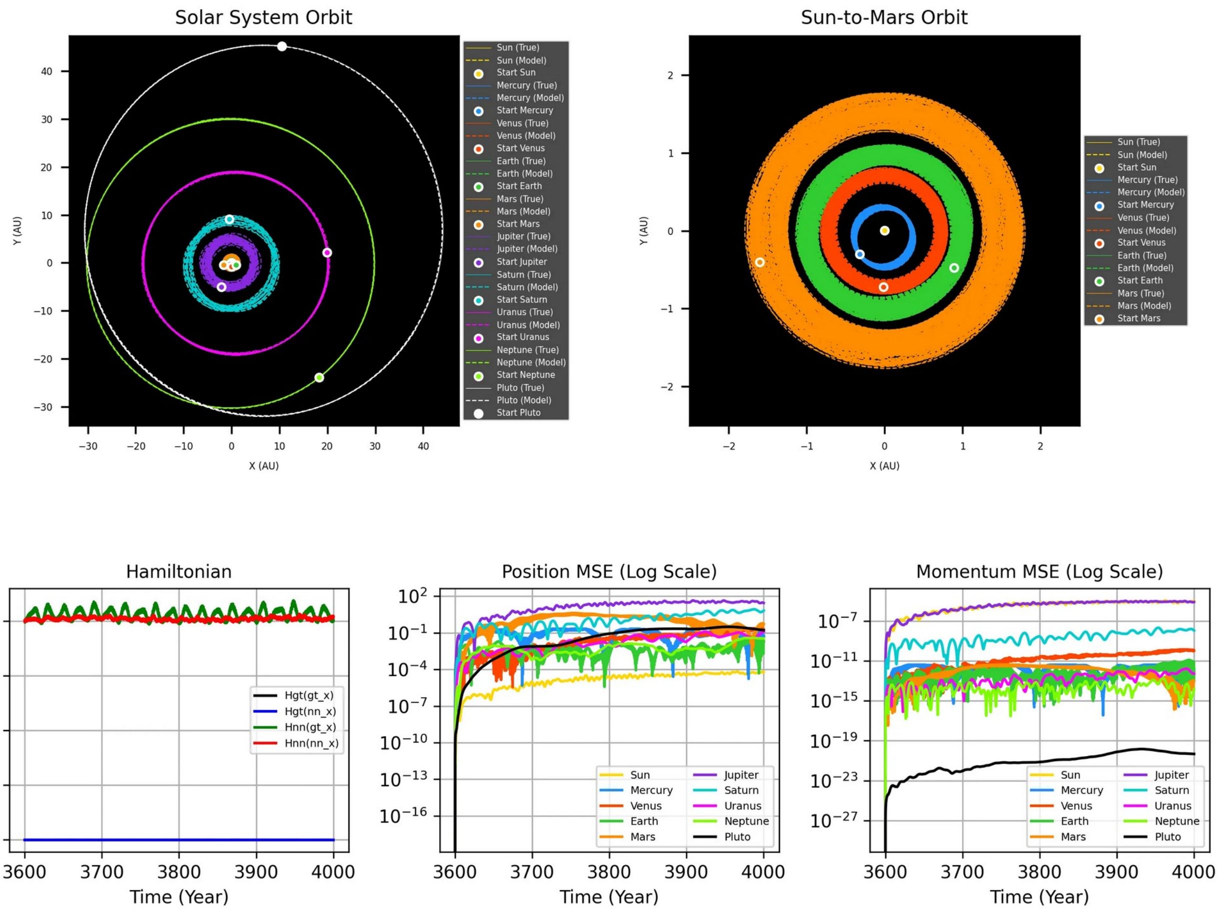


Figure 24: 400 years of orbit predictions and metrics for Block (Sparse) Plausible HNN (BLK-P-HNN) at the optimal configuration (layers l=4 and factor m=7) at evaluation partition 2 ranging from 3600 to 4000 years with Δt=0.01.

4.4. Limitations of BLK-P-HNN

As the integration horizon expands, all neural-network-driven ordinary differential equation (ODE) integrators inevitably exhibit a distinct numerical precession phenomenon. Over centuries, this structural artifact manifests visually as a progressive broadening of planetary orbits into defined, band-like trajectories. This secular drift is an expected consequence of accumulated local truncation errors and infinitesimal gradient mismatches inherent to functional neural approximations over extended temporal scales.

Despite this inevitable band-widening effect, a stark contrast emerges regarding system-level stability, defined here as the sustainable orbital lifespan. While the conventional HNN governed by Vector-wise Multiscale V2 normalization suffers from catastrophic unphysical drift—leading to the rapid ejection of inner planets and complete structural collapse within a 10-year threshold—the proposed BLK-P-HNN maintains a strictly bound state for the entire 400-year integration duration. Under the BLK-P-HNN framework, the celestial bodies experience bounded numerical precession but remain gravitationally locked within their realistic boundaries, never escaping the underlying Solar System manifold.

Nevertheless, a comparative analysis reveals that the average trajectory tracking performance of the dense P-HNN marginally outperforms the sparse BLK-P-HNN, as illustrated in Figures A1 and A2. When focusing exclusively on the XY-projections of the orbits, the BLK-P-HNN exhibits seemingly tighter, more localized spatial patterns. However, empirical inspection indicates that this superficial XY-localization is achieved at the expense of elevated orbital dispersions along the Z-axis. In contrast, the dense P-HNN demonstrates a more isotropic error distribution, where the trajectory dispersion in the vertical Z-direction does not exceed the corresponding error profiles observed within the horizontal X and Y dimensions.

4.5. Empirical Observations on BLK-P-HNN and Even Activations

Interestingly, in the architectures split into a kinetic net (T-net) and a potential net (V-net), we observed that the components of the attention weight vector become all 1/2 when trained. This empirical phenomenon seemingly indicates that the neural network (NN) tends to interpret the ODE for the state variable in the scaled space as:

$$\dot{\hat{q}}=\hat{p}, \quad \dot{\hat{p}}=-\frac{\partial V}{\partial \hat{q}},$$

which structurally mirrors a canonical coordinate system. This intriguing behavior is consistently observed both in P-HNN and in BLK-P-HNN.

The positive potential energy that the V-net yields indicates that when we design the activation with an even function, the V-net tends to interpret the Solar system as a system similar to a complex system of coupled oscillators. The original design drive toward an even activation was the fact that the real potential energy is of the form of $V(\boldsymbol{r}_i,\boldsymbol{r}_j)=V(\sqrt{\boldsymbol{R}_{ij}^2})$, where $\boldsymbol{R}_{ij}=\boldsymbol{r}_i-\boldsymbol{r}_j$. In the approximation of V in terms of a power series expansion (i.e., Taylor series), only even terms in $\boldsymbol{R}_{ij}^{2k}$ $(k\geq 0)$ survive due to spatial symmetry.

Since an NN can be viewed as a universal function approximator, we expected the V-net to learn this pair-wise symmetry upon the even activation when given some linear combinations of $\hat{\boldsymbol{q}}_i$ and $\hat{\boldsymbol{q}}_j$ via the *Topology Adaptor.* Crucially, we conjectured that in order to properly mimic the rich and diverse combinations of coefficients for these higher-order O(2*k*) terms, a lateral amplification of neuronal numbers (expanding layer width) would be highly desirable and mathematically effective, rather than naively stacking deeper layers. This structural philosophy led us to the development of the P-HNN and subsequently to the BLK-P-HNN architecture. The results indicate that a long-term stable gravitational N-body system like our Solar system may be otherwise viewed as coupled complex oscillators under the potential of some constant value plus even terms in the relative distances—a phenomenological perspective that seems to align smoothly with the intrinsic representations learned by both the P-HNN and BLK-P-HNN frameworks.

Additionally, we examined the learned weights of the attention aggregation layer within the BLK-P-HNN. This attention mechanism was introduced with the intention of guiding the network to learn coefficients that correspond to the gravitational potential constants $Gm_i m_j$ within the Multiscale V2 scaled space ($\hat{\boldsymbol{q}}_i = \boldsymbol{q}_i / max_{sample} \| \boldsymbol{q}_{sample,i} \|$). When comparing the parameters of the trained model against the logarithmic values of the true, unscaled gravitational coefficients, $\ln(Gm_i m_j)$, a complete match was not achieved; however, a partial correlation was observed (**Figure 25**). Interestingly, the network automatically assigned the largest weights to pairs involving the Sun, and it captured **l**imited similar patterns for interactions involving high-mass bodies such as Jupiter. This partial alignment suggests that the block sparse structure provides a baseline incentive for the model to explore physically meaningful parameter spaces, even though discrepancies remain.

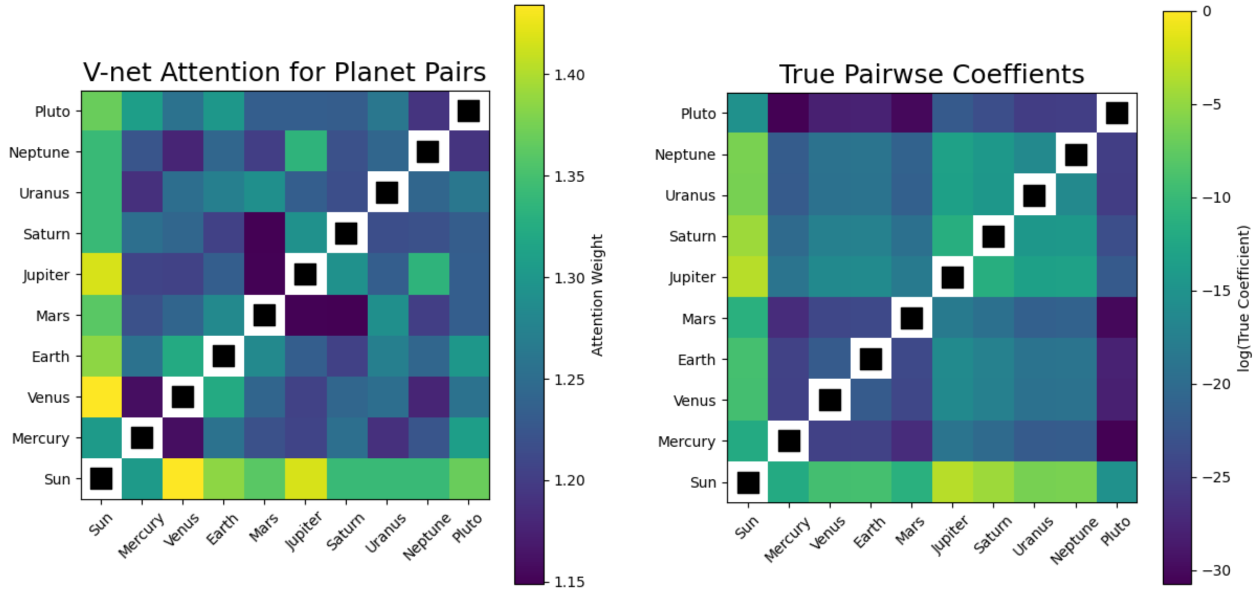


***Figure 25: Comparison of V-net Attention and True Pair-wise Potential**: (a) attention in V-net of BLK-P-HNN, (b) true $Gm_i m_j$ coefficients of the pair gravity potentials in the raw space.*

4.6. Empirical Analysis of the Separable Latent Linear Hamiltonian Neural Network Architecture

The architecture was configured utilizing $\log(x \tanh(x)+1)$ as the activation mapping. The resulting numerical integration and its physical implications on the preservation of the system's dynamics are detailed in **Figure 26**.

1) Multiple Normalization
To mitigate the magnitude disparities inherent in the kinematics and dynamics of a multi-body gravitational system, and to ensure stable convergence during optimization, a multiple normalization scheme at the degree of freedom level was implemented. Unlike global normalizations that can dilute the influence of bodies with minor spatial variations, this approach preserves the relative scale of each dimension.

Each component of the physical position and momentum vectors is independently scaled using its maximum absolute value recorded in the dataset, producing the normalized inputs $\hat{q}$ and $\hat{p}$.

Mathematically, the inverse transformation (from the normalized space to the physical space) is defined by the Hadamard product (element-wise):

$$q = \hat{q} \odot q_{max}, \qquad p = \hat{p} \odot p_{max},$$

where q_{max} and p_{max} are the 30-element scaling vectors, defined as $\max(|q|)$ and $\max(|p|)$, respectively. This element-wise scaling directly corresponds to the Multiscale V2 normalization scheme introduced in Eq. (4). Bounding the neural network inputs within the $[-1,1]$ range optimizes the gradient flow without distorting the physical geometry of the problem. The extracted scaling factors vary depending on the temporal resolution of the simulation. The q_{max} and p_{max} vectors obtained for the respective training regimes and integration steps are detailed below.

2) **Experimentation on Temporal Mesh Resolution**

The network's performance was evaluated across two temporal mesh resolutions. For the fine mesh (Δt=0.0005), the potential energy MLP was configured with 3 hidden layers of 256 neurons each and trained over 100 epochs. For the coarse mesh (Δt=0.01), the architecture was adjusted to 2 hidden layers of 256 neurons each and trained over 1000 epochs. The models were trained using simulation data from year 0 to 3200, and their dynamic stability was evaluated over the integration period from year 3200 to 4000.

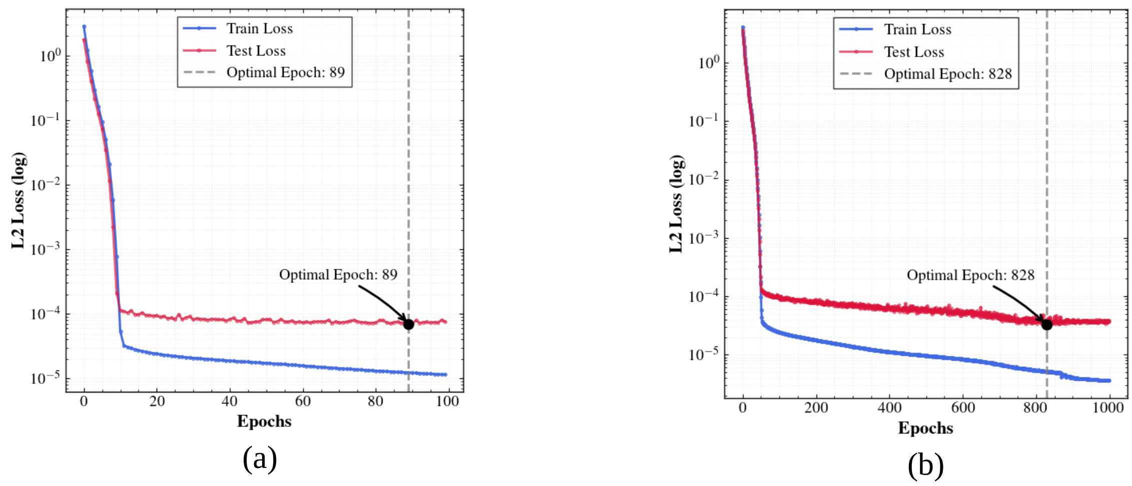


***Figure 26: Convergence profiles of the SLL-HNN architecture**: Red dashed lines indicate the best testing set performance epoch: (a) fine temporal mesh (Δt=0.0005), epoch 89; (b) coarse temporal mesh (Δt=0.01), epoch 828.*

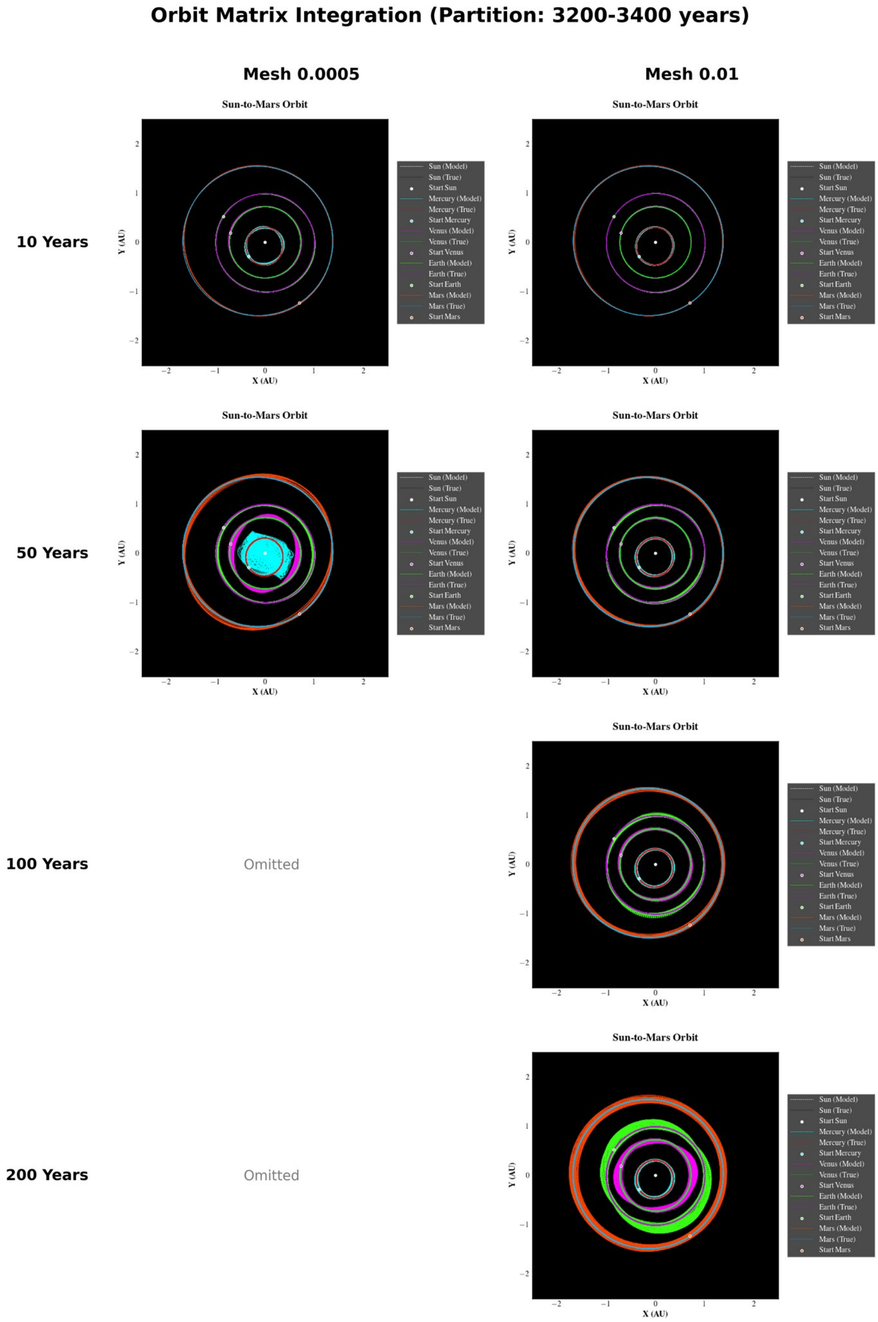


Figure 27: Orbital integration of the inner domain (Sun to Mars): Comparison between the model trained with a fine mesh (Δt=0.0005, left column) and a coarse mesh (Δt=0.01, right column). *Rows correspond to evaluation years 10, 50, 100, and 200 (absolute simulation years 3210, 3250, 3300, and 3400). Panels corresponding to years 100 and 200 for the fine mesh configuration were omitted due to numerical divergence.*

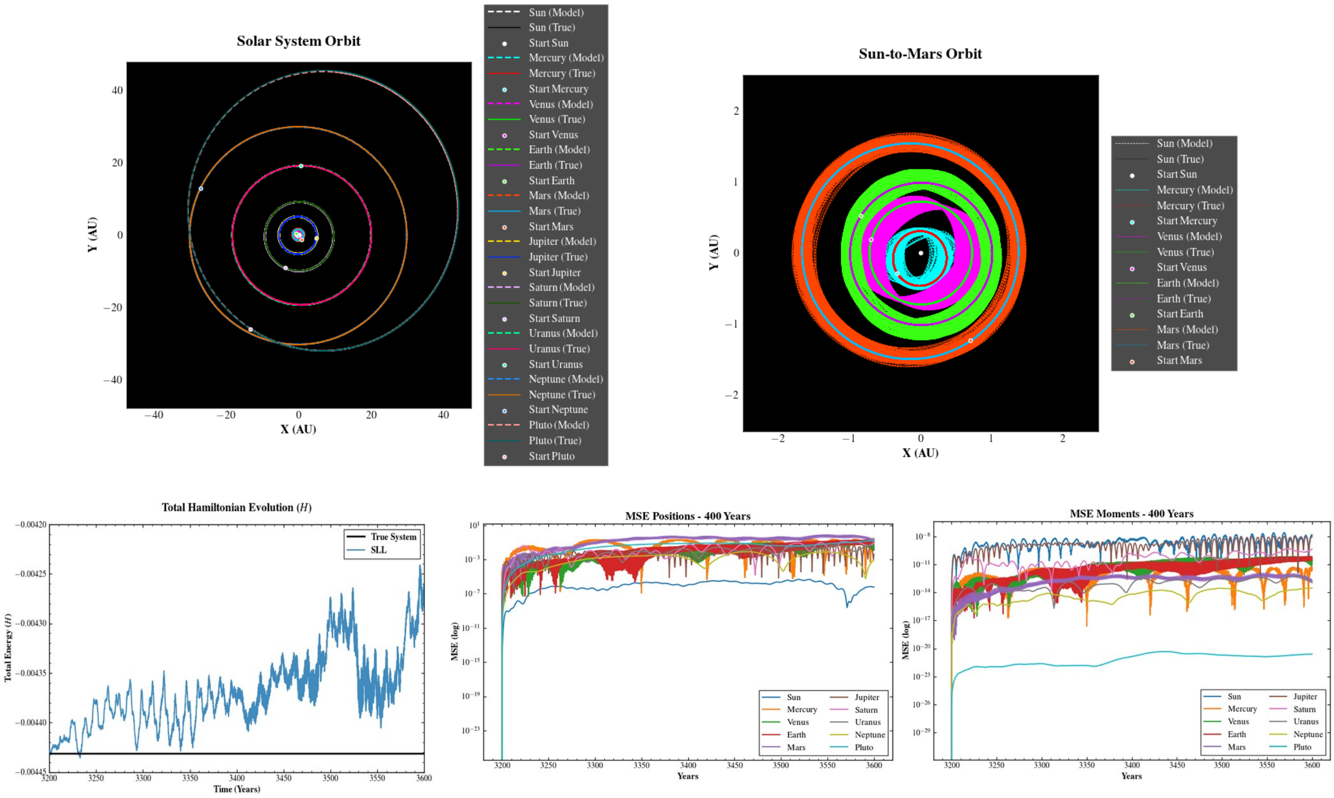


Figure 28: Kinematic and dynamic evolution of the coarse mesh architecture Δt=0.01. *The panels detail the complete and inner orbital trajectories, the predicted total Hamiltonian (H), and the Mean Squared Error (MSE) for position (q) and momentum (p).*

The model structured with the coarse temporal mesh (Δt=0.01, 2 hidden layers) demonstrated superior dynamic performance compared to the fine mesh configuration (Δt=0.0005, 3 hidden layers). The fine mesh architecture registered numerical divergence at evaluation year 50 due to the loss of confinement in Pluto's orbit. In contrast, the coarse mesh model maintained system stability until reaching its physical integration limit at evaluation year 200.

Figure 28 presents the kinematic and dynamic evolution of the coarse mesh architecture (Δt=0.01). The evaluation details the complete and inner orbital trajectories, the predicted energy output, and the Mean Squared Error (MSE) for position (q) and momentum (p). The model preserves the bound state of the celestial bodies without unphysical ejections. However, substantial numerical drift accumulates, generating orbital noise as the integration progresses.

The physical limitation established at the 200-year threshold is quantitatively supported by the energy evolution detailed in **Figure 28**. Midway through the 400-year evaluation window, the total Hamiltonian (H) registers a drastic increase in the conservation error. This severe accumulation of numerical drift translates directly into a topological failure of the innermost orbit**;** Mercury's trajectory collapses and widens, progressively intruding into the bounded regions of adjacent planets. This specific phenomenon of orbital collapse characterized by the complete filling of the interior geometric perimeter matches the instability patterns documented in the baseline P-HNN and BLK-P-HNN frameworks, corroborating the absolute physical boundary of the proposed architecture.

5.Summary and Conclusions

In this work, we have systematically investigated the scaling boundaries of conventional black-box HNNs applied to high-dynamic-range, multi-body gravitational systems. While standard architectures exhibit severe gradient stiffness or long-term energetic failures leading to domino-effect planetary ejections, our integration of Vector-wise Multiscale V2 normalization and explicit (T-V) energy separation via the P-HNN extends the sustainable orbital lifespan past the critical 248-year orbital period of Pluto, effectively mitigating macro-scale system decay. To manage the parameter inflation and computational tractability constraints associated with long-term high-fidelity trajectories, we developed the Block (Sparse) Plausible Hamiltonian Neural Network (BLK-P-HNN). By leveraging a batch matrix multiplication-based Topology Adaptor to enforce structural sparsity, the BLK-P-HNN achieves a 400-fold parameter compression (from 88.6 MB to 200 kB). Empirical validation throughout a 400-year horizon reveals an informative performance trade-off: while the extreme structural compression introduces subtle localized orbital variations and an increased error dispersion concentrated along the vertical Z-axis compared to the dense model, the sparse framework successfully maintains a gravitationally bound state without catastrophic orbital breakdown.

Beyond structural compression, empirical observations of the trained frameworks provide useful insights into neural representations of physical symmetries. By deploying even activation functions, the V-net was explicitly designed under the expectation that the network would learn the pairwise spatial symmetries found in the Taylor series expansions of gravitational potentials. Furthermore, based on our mathematical reasoning, we investigated the structural impact of width versus depth; while structural depth variations were constrained to a comparative evaluation between three- and four-layer configurations, scaling the features laterally demonstrated that expanding the hidden layer width up to the boundary of over-parameterization serves as an effective approach to capture the diverse combinations of higher-order potential coefficients. Ultimately, these findings demonstrate that incorporating physical inductive biases with structured sparsity offers a viable, informative pathway to balance extreme model compression with long-term macro-scale stability in complex multi-body celestial simulations.

The evaluation of the Single Linear Layer Hamiltonian Neural Network (SLL-HNN) assesses the viability of component-wise linear mappings coupled with dense representations. Although the framework preserves a gravitationally bound state over a 400-year integration window, it exhibits strict physical limitations. The SLL-HNN accumulates severe numerical drift and orbital noise, underperforming the baseline P-HNN in accuracy. Structurally, its 600 kB footprint (74,195 parameters) lacks the parameter efficiency of the 200 kB BLK-P-HNN. Consequently, the SLL-HNN operates strictly as an architectural proof-of-concept. It validates the geometric requirement of even mappings for macroscopic stability but remains quantitatively inferior to the BLK-P-HNN in the balance of computational cost and precision. The reported orbital instability confirms that restricting the physical-to-latent embedding to a strictly linear mapping inherently bottlenecks the model's representational capacity. Future work is required to systematically investigate the exact nature of this bottleneck. Subsequent research must explore alternative architectural configurations and conduct rigorous empirical testing to determine if the long-term numerical drift can be mitigated to extend the dynamic stability beyond the 400-year threshold.

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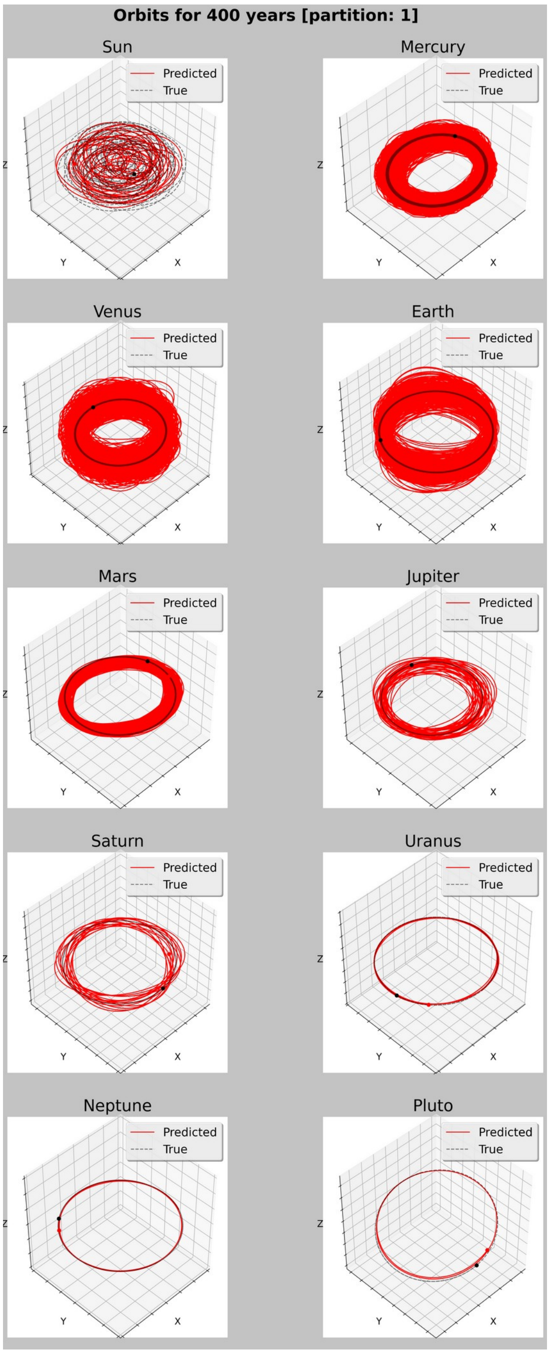
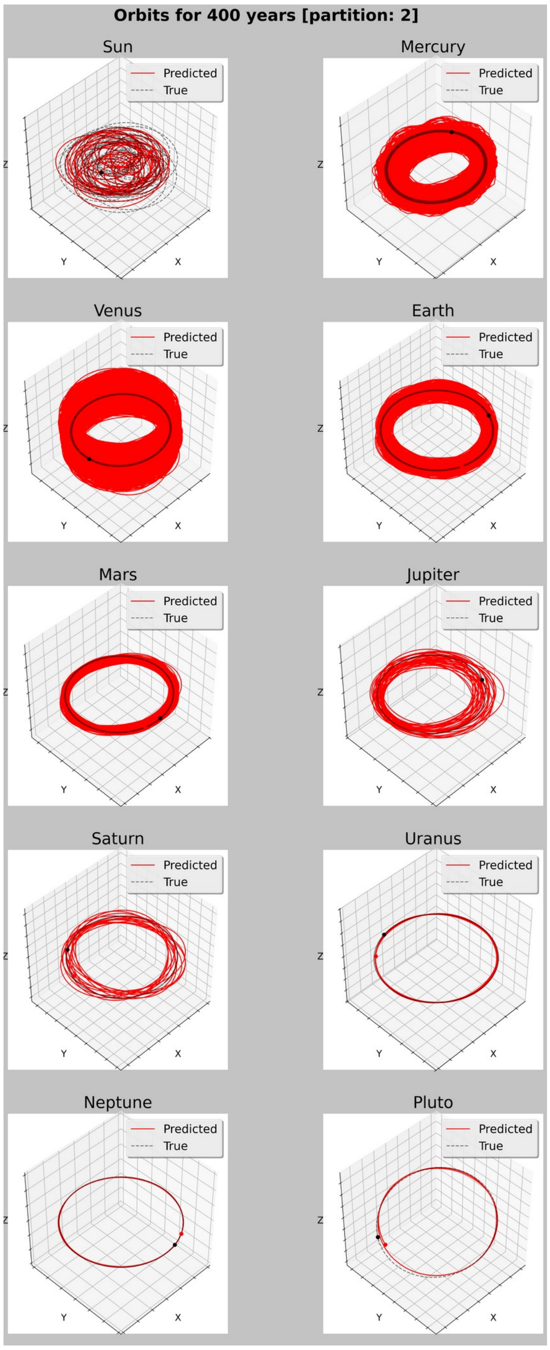
Appendices



(a)



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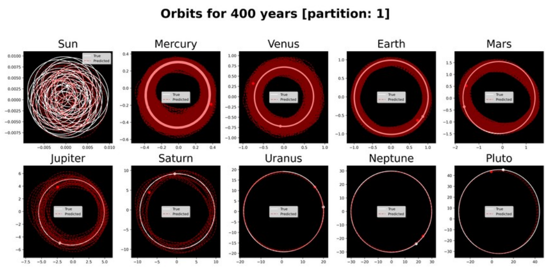

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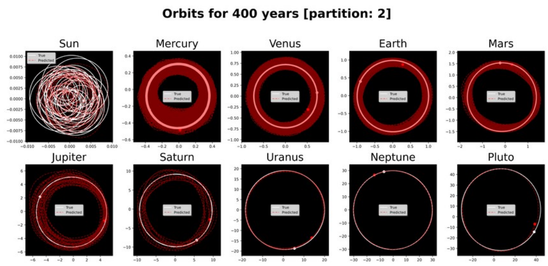
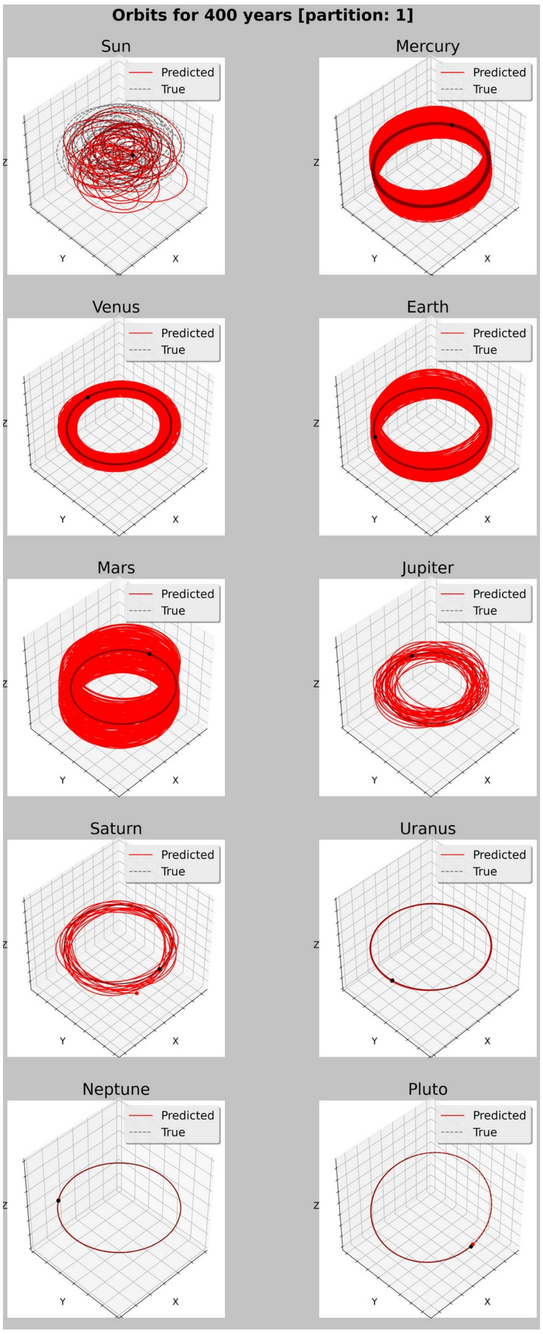
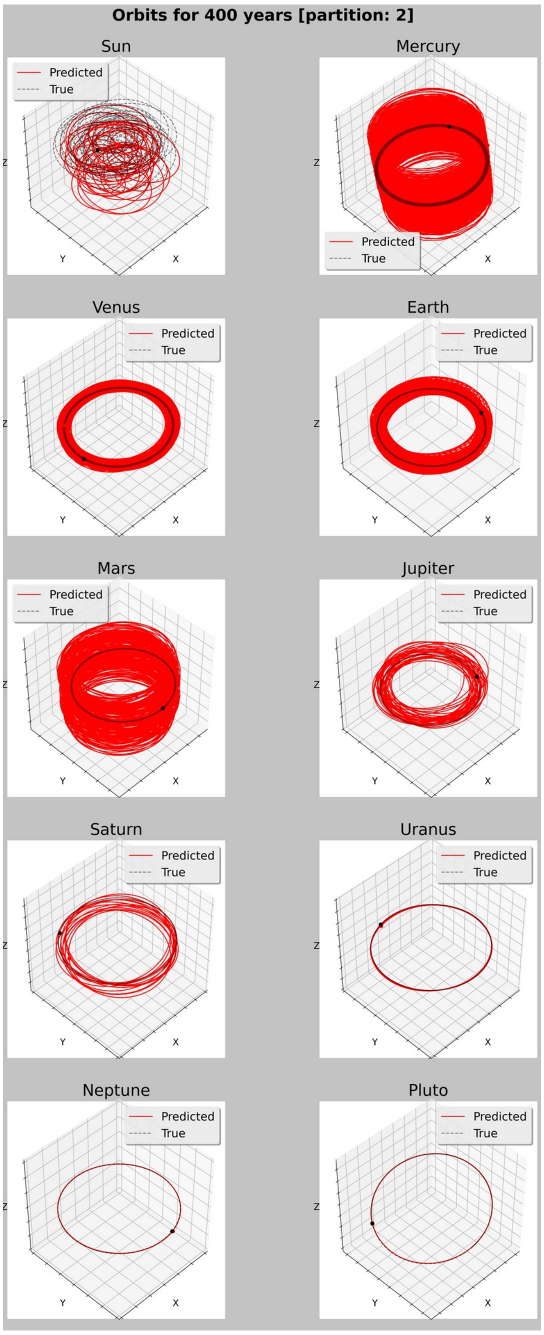


(d)

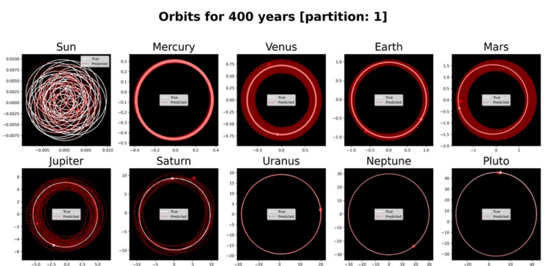
Figure A1: Orbit predictions per body with P-HNN(l=4, m=10) in Δt=0.01. *(a) and (b) are 3D, (c) and (d) are 2D orbits, for partitons 1 and 2, each, where partition 1 ranging from 3200 to 3600 years and partition 2 ranging from 3600 to 4000 years.*



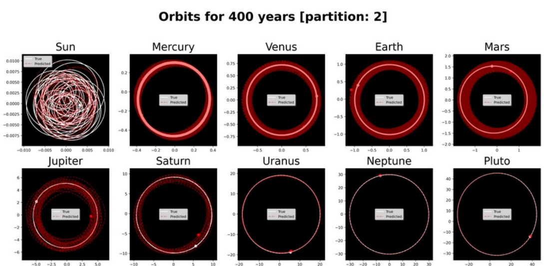
(a)



(b)



(c)



(d)

Figure A2: Orbit predictions per body with BLK-P-HNN(l=4, m=7) in Δt=0.01. *(a) and (b) are 3D, (c) and (d) are 2D orbits, for partitons 1 and 2, each, where partition 1 ranging from 3200 to 3600 years and partition 2 ranging from 3600 to 4000 years.*

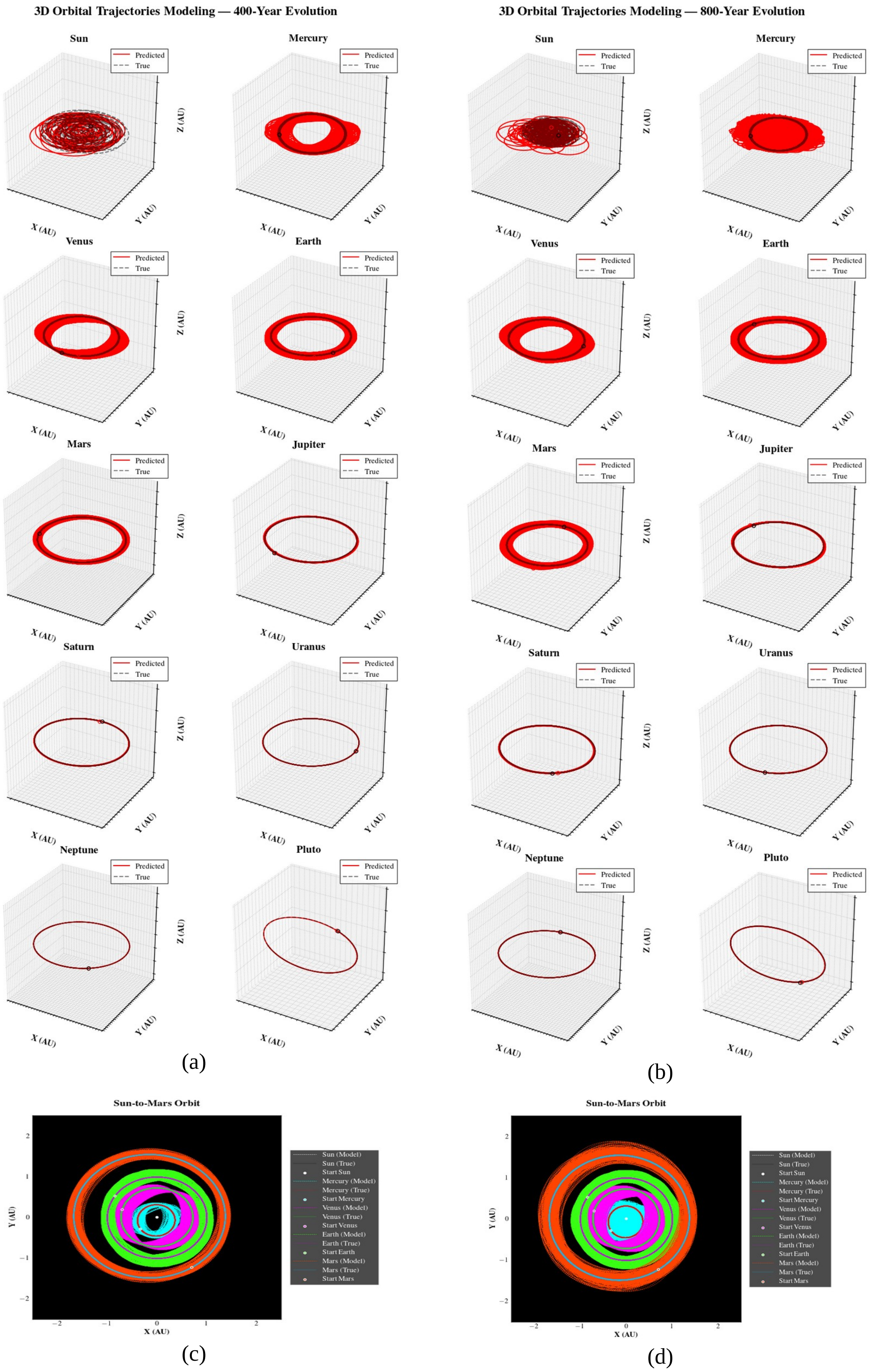


Figure A3: Body orbit predictions with SLL-HNN(l=2) at Δt=0.01. *(a) and (b) are 3D orbits, (c) and (d) are internal orbits of the Sun to Mars, for partitions 1 and 2, each, where partition 1 spans from 3200 to 3600 years and partition 2 spans from 3200 to 4000 years.*